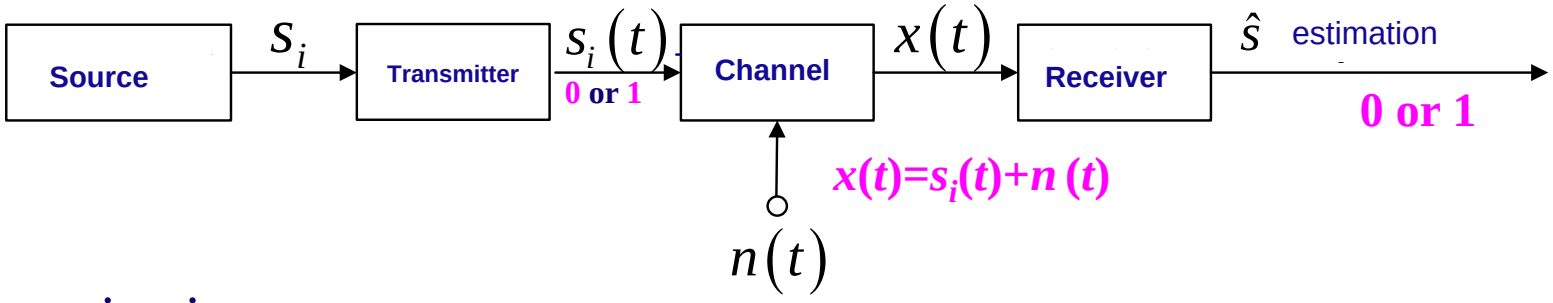
Signal Space Analysis and Optimal Receiver Design

1. **Signal space analysis**
2. Optimal receiver problem and MAP criteria
3. ML criteria and correlation receivers
4. Matched filter optimum receiver
5. Error performance analysis



Digital communication system

- Transmitter： *M* kinds of messages (waveform)
- Receiver： judge the noisy observed signal as some kind of sending message

——There is a risk of judgment

Optimal receiver problem： design the receiver to minimize the probability of decision errors

- Optimal receiver criteria, Optimal receiver design, performance
- Signal space analysis: geometric representation for visual analysis

GUIDE TO STUDY

- Vector space and signal space
- Digital signal vector representation
- Received signal vector representation

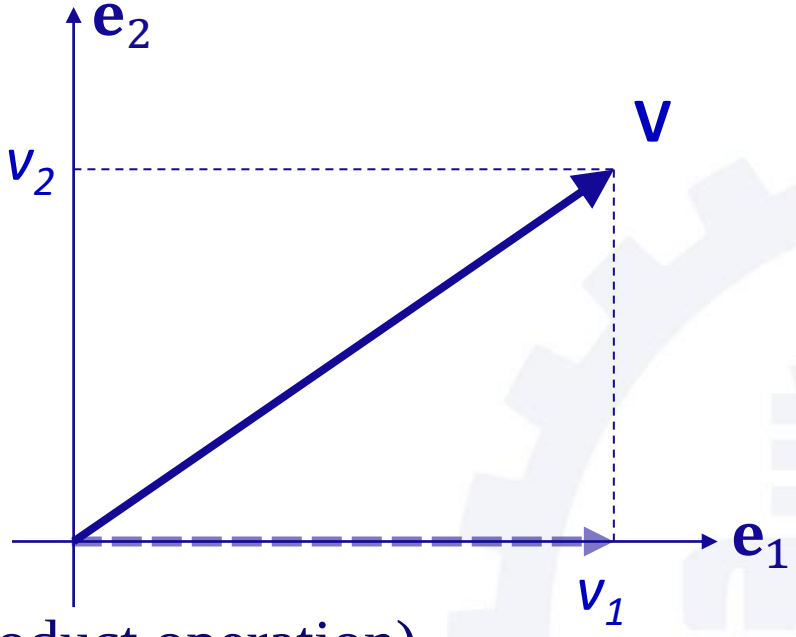
7.1 SIGNAL SPACE ANALYSIS

7.1.1 VECTOR SPACE AND SIGNAL SPACE

Vector Space

- N mutually orthonormal vector groups ($\boldsymbol{e_1}$, $\boldsymbol{e_2}$, …, $\boldsymbol{e_N}$) form an N-dimensional space, any vector V can be expressed as a linear sum of projected vectors on N coordinate axes

$$\mathbf{V} = \sum_{i=1}^{N} v_i \cdot \mathbf{e_i} \qquad \mathbf{e_i} \cdot \mathbf{e_j} = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$



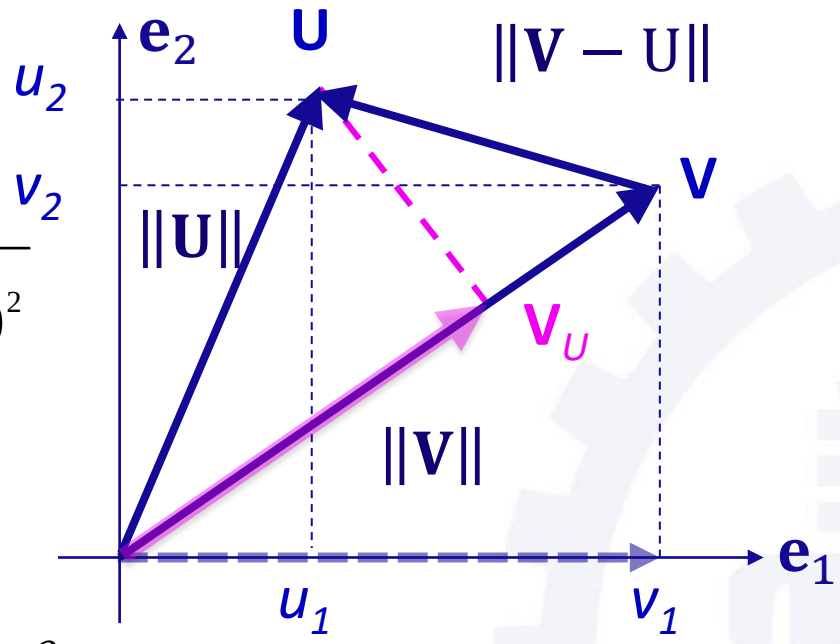
- $\mathbf{e_i}$: unit vector, orthogonal to each other
- ($\mathbf{e_1}$, $\mathbf{e_2}$, …, $\mathbf{e_N}$) normalized orthogonal basis (vector) in vector space
- v_i : projection of vector V onto $\mathbf{e_i}$ (inner product operation)
- $\mathbf{V}_i = v_i \cdot \mathbf{e_i}$: projection vector of $\mathbf{V}$ onto $\mathbf{e_i}$

Norm, distance and inner product

- Vector $\mathbf{U} = [u_1, u_2, \cdots, u_N]$ $\mathbf{V} = [v_1, v_2, \cdots, v_N]$
- Norm
  Vector length $\|\mathbf{U}\| = \sqrt{\sum_{i=1}^{N} u_i^2}$ $\|\mathbf{V}\| = \sqrt{\sum_{i=1}^{N} v_i^2}$
- Euclidean distance
  Length of difference vector of two vectors $\|\mathbf{U} - \mathbf{V}\| = \sqrt{\sum_{i=1}^{N} (u_i - v_i)^2}$
- Inner product
  The similarity of two vectors can also be understood as the product of the projection of one vector on another vector and the length of the vector

$$\mathbf{U} \cdot \mathbf{V} = \sum_{i=1}^{N} u_i v_i$$

$$\mathbf{U} \cdot \mathbf{V} = \|\mathbf{U}\|\|\mathbf{V}\|\cos\theta$$
$$= \|\mathbf{U}\| V_U = \|\mathbf{V}\| U_V$$



Signal space

- A set of signals $\{f_i(t),\ i=1,\ldots,N\}$

$$\int_{-\infty}^{\infty} f_k(t)\, f_l(t)\, dt = \begin{cases} 0, & k \neq l \\ 1, & k = l \end{cases}$$

Orthogonal
Energy=1

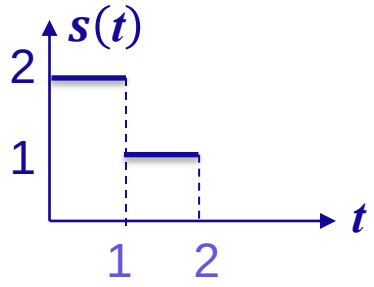
- $\{f_i(t)\}$ spanned *N dimensional signal space*

$$\Omega = \left\{ s(t) : s(t) = \sum_{i=1}^{N} w_i f_i(t),\ w_i \in R,\ R\text{为 real} \right\}$$

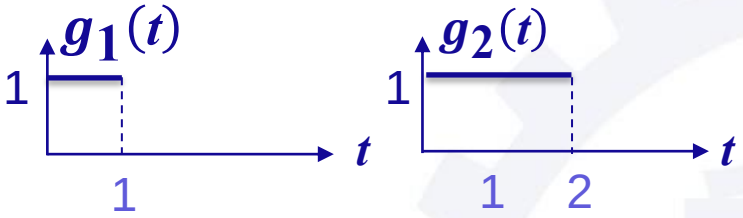
$\{f_i(t)\}$: orthonormal basis of signal space Ω

- Complete space: any signal $s(t) \in \Omega$ can always be expressed by the sum of the signals in the signal space

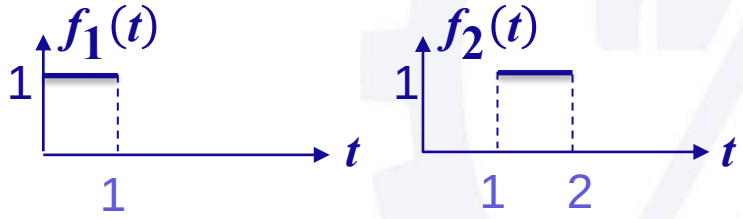
$$s(t) = \sum_{i=1}^{N} s_i \cdot f_i(t)$$



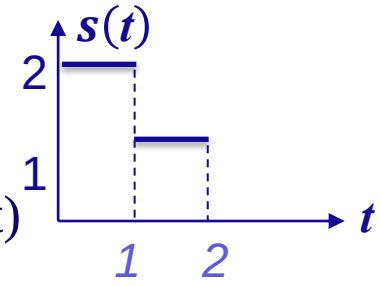
$$s(t) = g_1(t) + g_2(t)$$



$$s(t) = 2f_1(t) + f_2(t)$$



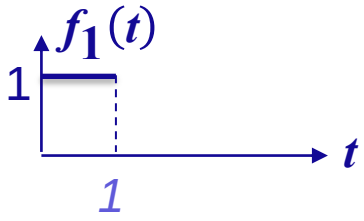
Signal vector representation

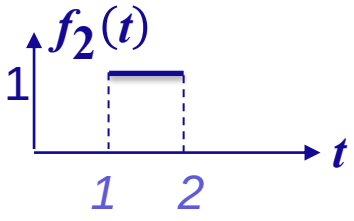


$$s(t) = 2f_1(t) + f_2(t)$$

- Base signal decomposition of signal s (t)

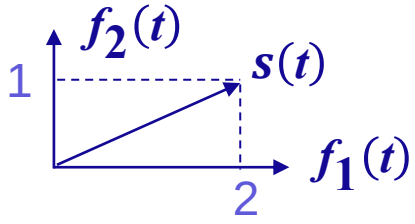
$$s(t) = \sum_{i=1}^{N} s_i \cdot f_i(t)$$





- Signal s(t) is a point in the orthogonal signal space, that is, a vector

$$\mathbf{s} = [s_1, s_2, \cdots, s_N]$$

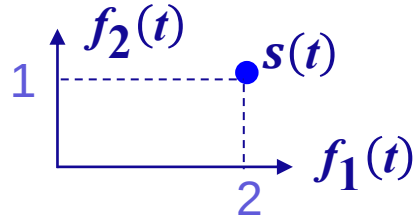


- s_i is projection of $s(t)$ on $f_i(t)$

$$s_i = \int_{-\infty}^{\infty} s(t) \cdot f_i(t) dt$$

- The energy of the signal s(t) corresponds to the square of the vector norm

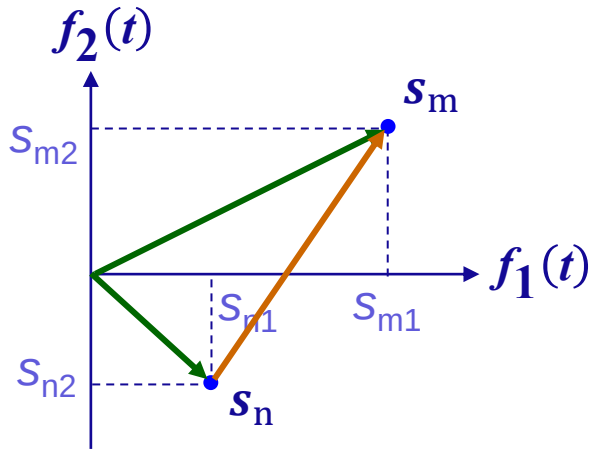
$$E_s = \int_{-\infty}^{\infty} s^2(t) dt = \sum_{i=1}^{N} s_i^2 = \|\mathbf{s}\|^2$$



Correlation coefficient and signal distance

$$s_m(t) \qquad s_m = [s_{m1}, s_{m2}, \cdots, s_{mN}]$$

$$s_n(t) \qquad s_n = [s_{n1}, s_{n2}, \cdots, s_{nN}]$$



- Euclidean distance between signals

$$d_{mn} = \|s_m - s_n\| = \sqrt{\sum_i (s_{mi} - s_{ni})^2}$$

$$= \sqrt{E_m + E_n - 2\sqrt{E_m E_n}\rho_{mn}}$$

$$= \sqrt{\int_{-\infty}^{\infty} [s_m(t) - s_n(t)]^2 \, dt}$$

- *when* $E_m = E_n = E$ $\quad d_{mn} = \sqrt{2E(1-\rho)}$

- Correlation coefficient between signals
  - Normalized vector inner product
  - $\rho \in [-1, 1]$

$$\rho_{mn} = \frac{1}{\sqrt{E_m E_k}} \int_{-\infty}^{\infty} s_m(t) s_n(t) dt$$

$$= \frac{s_m \cdot s_n}{\sqrt{E_m E_k}} = \frac{s_m \cdot s_n}{\|s_m\| \|s_n\|}$$

Signal orthogonalization method

Gram-Schmidt method

Input: Given a signal set over $[t_0, t_0+T_s]$

$\{s_1(t), s_2(t), \ldots, s_M(t)\}$

Output: Its orthogonal basis function $\{f_1(t), f_2(t), \ldots, f_N(t)\}$

Steps:

(1)$v_1(t) = s_1(t)$, obtain a basis function $f_1(t) = v_1(t)/\|v_1\|$

(2)$v_2(t) = s_2(t) - s_{21}f_1(t)$, s_{21} is projection of $s_2(t)$ on $f_1(t)$, obtain a basis function $f_2(t) = v_2(t)/\|v_2\|$

(3)$v_3(t) = s_3(t) - s_{32}f_2(t) - s_{31}f_1(t)$,
obtain a basis function $f_3(t) = v_3(t)/\|v_3\|$

(4) Continue the above steps until all $s_i(t)$ have been used.
The resulting non-zero energy $f_i(t)$ form a set of orthogonal basis signals of $N \leq M$ dimensions

Examples of signal orthogonalization methods

ASK signal

$$\begin{cases} s_1(t) = A\cos\omega_0 t, & 1 \\ s_2(t) = 0, & 0 \end{cases} \qquad 0 \le t \le T_s$$

$s_1(t)$ signal energy

$$E_1 = \int_0^{T_s} s_1^2(t)\,dt = \int_0^{T_s} A^2\cos^2\omega_0 t\,dt = \frac{A^2 T_s}{2}$$

Orthogonal basis function

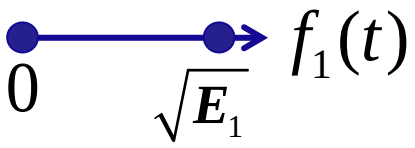
$$f_1(t) = \frac{s_1(t)}{\|s_1(t)\|} = \frac{s_1(t)}{\sqrt{E_1}} = \sqrt{\frac{2}{T_s}}\cos\omega_0 t$$

Signal orthogonal representation

$$s_1(t) = \sqrt{E_1}\,f_1(t), \quad s_2(t) = 0, \quad 0 \le t \le T_s$$

Signal vector form

$$\mathbf{s}_1 = \left(\sqrt{E_1}\right), \ \mathbf{s}_2 = (0)$$



VECTOR SPACE AND SIGNAL SPACE SUMMARY

- Signal vector representation facilitates intuitive design of signal sets
- The signal norm corresponds to the signal energy, and the signal inner product corresponds to the signal correlation
- Signal distance is directly related to communication system performance

7.1.2 VECTOR REPRESENTATION OF DIGITAL SIGNALS

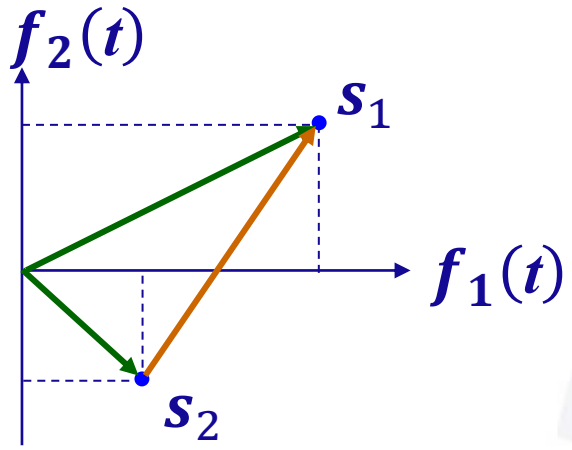
Constellation

Signal distance

Constellation

- Signal vector diagram
  - M-ary signal set: M points in N-dimensional orthogonal signal space
  - Signal: point, vector
  - Signal energy: norm squared

$$E_s = \int_{-\infty}^{\infty} s^2(t)dt = \sum_{i=1}^{N} s_i^2 = \|\mathbf{s}\|^2$$



  - Signal distance: length of difference vector

$$d_{12} = \sqrt{\int_{-\infty}^{\infty} [s_1(t) - s_2(t)]^2 dt}$$

Constellation diagram of digital frequency band signals

- Orthogonal base signal

$$f_1(t) = \frac{1}{\sqrt{T_s}} \qquad 0 \le t \le T_s$$

- Unipolar binary baseband signal

$$\begin{cases} s_1(t) = \sqrt{E}\, f_1(t) \\ s_2(t) = 0 \end{cases} \qquad d_{12} = \sqrt{E}$$

- Bipolar binary baseband signal

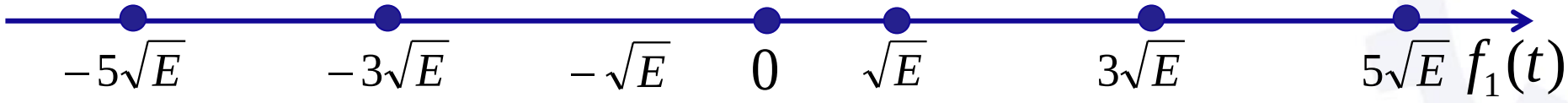
$$\begin{cases} s_1(t) = \sqrt{E}\, f_1(t) \\ s_2(t) = -\sqrt{E}\, f_1(t) \end{cases} \qquad d_{12} = 2\sqrt{E}$$

- M-ary baseband signal

$$s_i(t) = a_i \sqrt{E}\, f_1(t)$$

$$a_i = \pm 1, \pm 3, \pm 5, \cdots$$

$$d_{\min} = 2\sqrt{E} \qquad E: \textbf{Symbol energy}$$



Constellation diagram of digital frequency band signals: one-dimensional modulation

- Orthogonal base signal

$$f_1(t)=\sqrt{\frac{2}{T_s}}\cos\omega_0 t \qquad 0\le t\le T_s$$

- 2ASK

$$\begin{cases} s_1(t)=\sqrt{E}\,f_1(t) \\ s_2(t)=0 \end{cases} \qquad d_{12}=\sqrt{E}$$

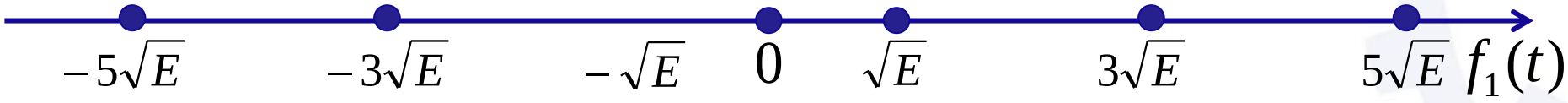
- 2PSK

$$\begin{cases} s_1(t)=\sqrt{E}\,f_1(t) \\ s_2(t)=-\sqrt{E}\,f_1(t) \end{cases} \qquad d_{12}=2\sqrt{E}$$

- MASK

$$s_i(t)=a_i\sqrt{E}\,f_1(t) \qquad d_{\min}=2\sqrt{E}$$

$$a_i=\pm1,\pm3,\pm5,\cdots$$



Constellation diagram of digital frequency band signals: two-dimensional modulation

- Orthogonal base signal

$$f_1(t)=\sqrt{\frac{2}{T_s}}\cos\omega_0 t \qquad f_2(t)=\sqrt{\frac{2}{T_s}}\sin\omega_0 t \qquad 0\le t\le T_s$$

- MPSK（M-element phase shift keying）

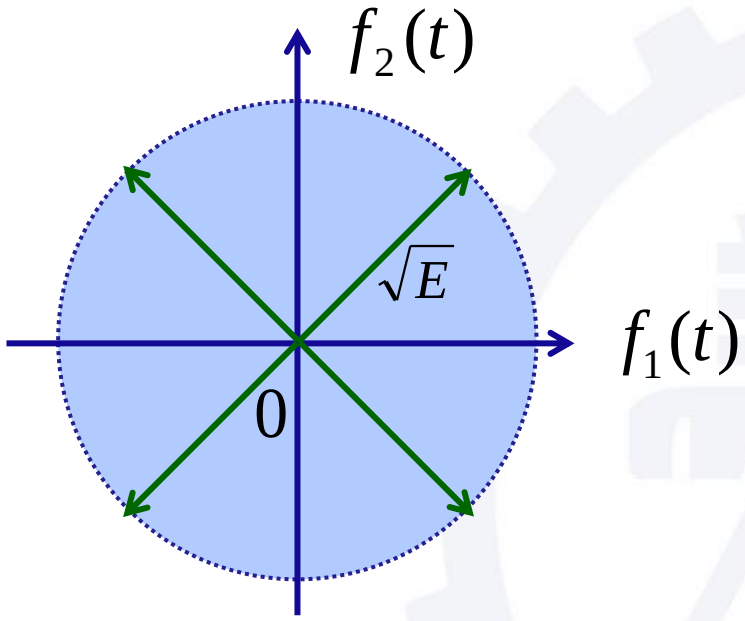
$$s_{MPSK}(t)=\sqrt{E}\cos\theta_i\cdot f_1(t)-\sqrt{E}\sin\theta_i\cdot f_2(t)$$

$$\mathbf{s}_i=\left[\sqrt{E}\cos\theta_i,-\sqrt{E}\sin\theta_i\right]$$

$$\|\mathbf{s}_i\|=\sqrt{E} \qquad \theta_i=\frac{(2i-1)\pi}{M},\ \ i=1,\cdots,M/2$$

$$\Delta\theta=\frac{2\pi}{M}$$

$$d_{\min}=2\sqrt{E}\sin\frac{\Delta\theta}{2}$$



$$\cos(\alpha+\beta)=\cos\alpha\cos\beta-\sin\alpha\sin\beta$$

Constellation diagram of digital frequency band signals: two-dimensional modulation

- Orthogonal base signal

$$f_1(t)=\sqrt{\frac{2}{T_s}}\cos\omega_0 t \qquad f_2(t)=\sqrt{\frac{2}{T_s}}\sin\omega_0 t \qquad 0\le t\le T_s$$

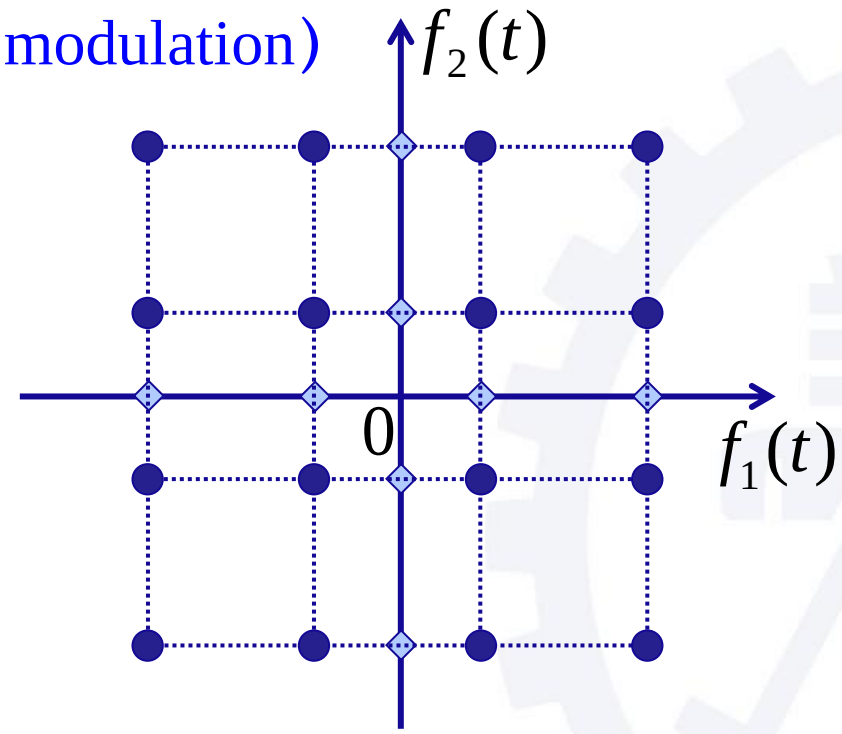
- MQAM（M-ary quadrature amplitude modulation）

$$s_{MQAM}(t)=X_i\cos\omega_0 t-Y_i\sin\omega_0 t$$

$$=X_i\sqrt{\frac{T_s}{2}}f_1(t)-Y_i\sqrt{\frac{T_s}{2}}f_2(t)$$

$$s_i=[X_i\sqrt{\frac{T_s}{2}},-Y_i\sqrt{\frac{T_s}{2}}]$$

- d_{min}=?



Constellation diagram of digital frequency band signals: two-dimensional modulation

- Orthogonal base signal

$$f_1(t)=\sqrt{\frac{2}{T_s}}\cos\omega_1 t \qquad f_2(t)=\sqrt{\frac{2}{T_s}}\cos\omega_2 t \qquad 0\le t\le T_s$$

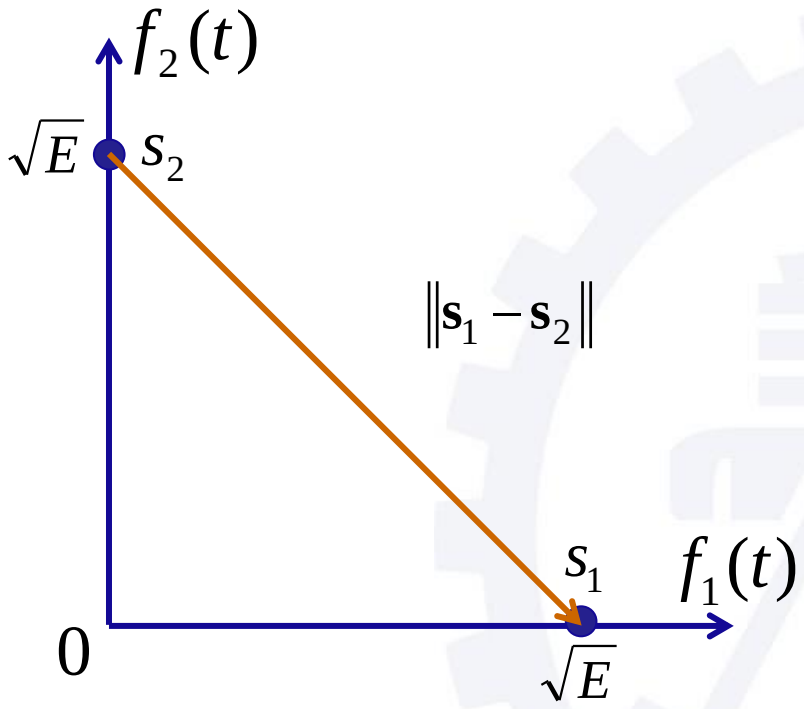
$$\int_{-\infty}^{\infty} f_1(t)\,f_2(t)dt=0$$

$$2\Delta f=f_2-f_1=\frac{n}{2}R_B=\frac{n}{2T_S}$$

- 2FSK

$$\begin{cases} s_1(t)=\sqrt{E}\,f_1(t) \\ s_2(t)=\sqrt{E}\,f_2(t) \end{cases} \qquad \begin{cases} \mathbf{s}_1=\left(\sqrt{E},0\right) \\ \mathbf{s}_2=\left(0,\sqrt{E}\right) \end{cases}$$

$$d_{12}=\sqrt{2E}$$



Constellation diagram of digital frequency band signals: N-dimensional modulation

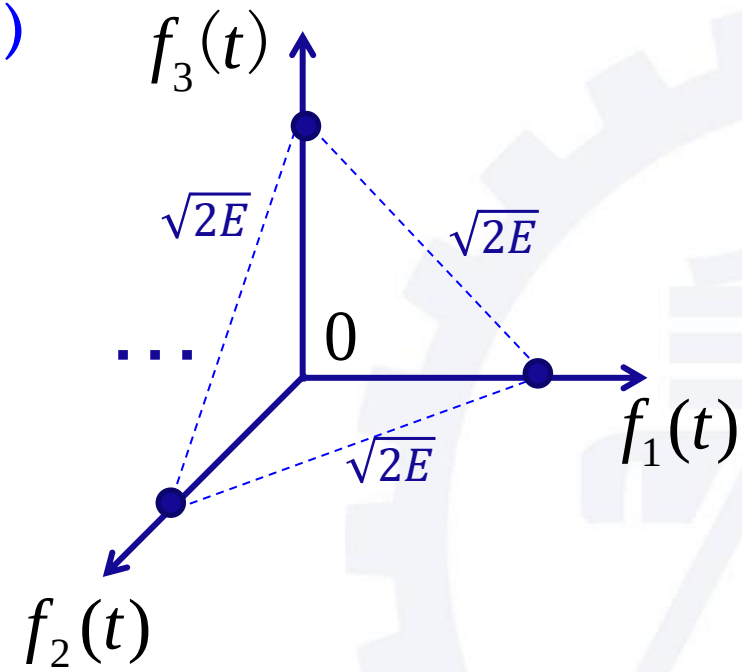
- Orthogonal base signal

$$f_i(t) = \sqrt{\frac{2}{T_s}} \cos \omega_i t, \quad i = 1, \ldots, N \quad 0 \le t \le T_s$$

$N \le M$

- MFSK （M-element frequency shift keying）

$$\begin{cases} s_1(t) = \sqrt{E} f_1(t) \\ s_2(t) = \sqrt{E} f_2(t) \\ \vdots \\ s_M(t) = \sqrt{E} f_M(t) \end{cases} \qquad \begin{cases} \mathbf{s}_1 = (\sqrt{E}, 0, \ldots, 0) \\ \mathbf{s}_2 = (0, \sqrt{E}, \ldots, 0) \\ \vdots \\ \mathbf{s}_M = (0, 0, \ldots, \sqrt{E}) \end{cases}$$



$$d_{ij} = \sqrt{2E} \quad i \ne j$$

DIGITAL SIGNAL VECTOR REPRESENTATION: SUMMARY

- Constellation diagram: geometric representation of a collection of digital signals
- One-dimensional modulation: M-ary baseband code, MASK, 2PSK
- Two-dimensional modulation: MPSK, QAM, 2FSK
- N-dimensional modulation: MFSK

7.1.3 RECEIVED SIGNAL VECTOR REPRESENTATION



The projection of received signal on the signal space

- Signal space orthogonal basis $\{f_i(t); i = 1, \cdots, N\}$
- Noise signal $n(t) = \sum_{i=1}^{N} n_i \cdot f_i(t) + v(t)$
- Received signal

$$x(t) = s(t) + n(t)$$

$$= s(t) + \sum_{i=1}^{N} n_i \cdot f_i(t) + v(t)$$

$$= \sum_{i=1}^{N} x_i \cdot f_i(t) + v(t)$$

$$\downarrow$$

$$\mathbf{x} = \mathbf{s} + \mathbf{n} = [x_1, x_2, \cdots, x_N]$$

$$\mathbf{s} = [s_1, s_2, \cdots, s_N]$$

$$\mathbf{n} = [n_1, n_2, \cdots, n_N]$$

$$x_i = s_i + n_i$$

$$n_i = \int_{-\infty}^{\infty} n(t) \cdot f_i(t) dt$$

$$v(t) = n(t) - \sum_{i=1}^{N} n_i \cdot f_i(t)$$

$$E\left[v(t) s_i(t)\right] = 0$$

v(t) Gaussian random variable

Independent of any n_i,
unrelated to any s_i

When comparing the distance between the received signal and the sample signal, effect of v (t) can be ignored

Distance between received signal and sample signal

If send $s_1(t)$

$$x(t) = s_1(t) + n_1(t)$$

$$d\left[x(t), s_1(t)\right] = \sqrt{\int_{-\infty}^{\infty} \left(x(t) - s_1(t)\right)^2 dt}$$

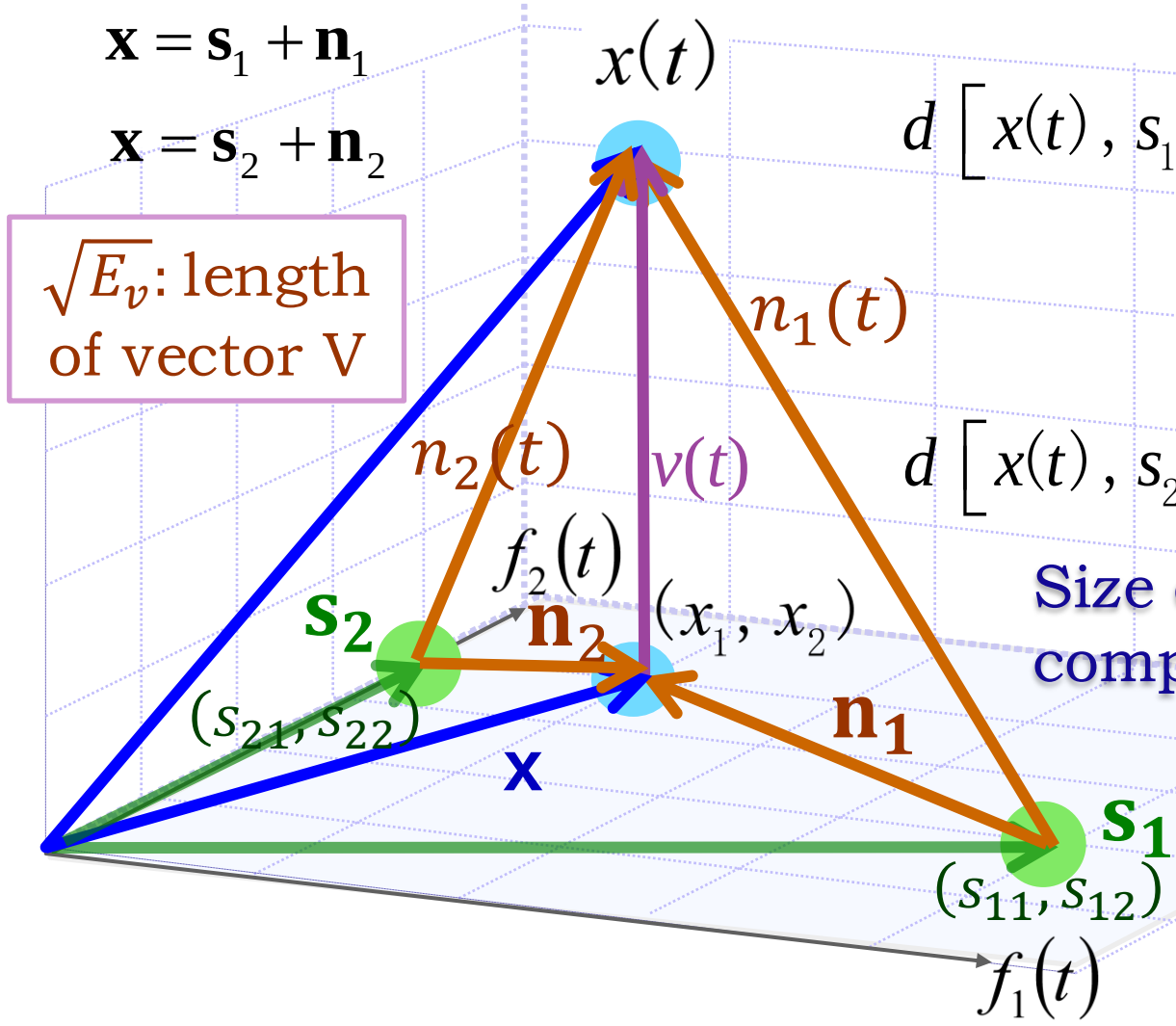
$$= \sqrt{|\mathbf{n}_1|^2 + E_v}$$

$$d\left[x(t), s_2(t)\right] = \sqrt{|\mathbf{n}_2|^2 + E_v}$$

$\mathbf{x} = \mathbf{s}_1 + \mathbf{n}_1$

$\mathbf{x} = \mathbf{s}_2 + \mathbf{n}_2$

$\sqrt{E_v}$: length of vector V



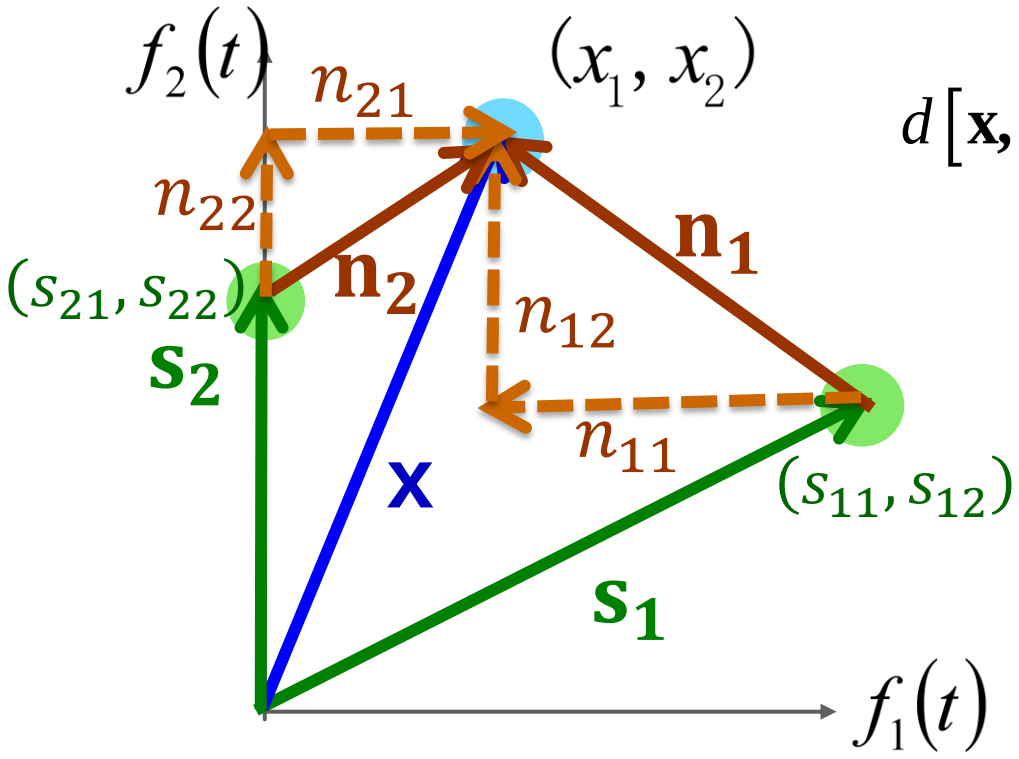
Size of v (t) has no effect on distance comparison, only need consider

distance between received signal and sample signal in signal space

Distance in signal space

$$\mathbf{x} = \mathbf{s}_1 + \mathbf{n}_1 = [x_1, x_2]$$

$$\mathbf{x} = \mathbf{s}_2 + \mathbf{n}_2 = [x_1, x_2]$$



$$d[\mathbf{x}, \mathbf{s}_1] = \sqrt{(x_1 - s_{11})^2 + (x_2 - s_{12})^2}$$
$$= \sqrt{n_{11}^2 + n_{12}^2}$$

$$d[\mathbf{x}, \mathbf{s}_2] = \sqrt{(x_1 - s_{21})^2 + (x_2 - s_{22})^2}$$
$$= \sqrt{n_{21}^2 + n_{22}^2}$$

length of the projected component of noise in the signal space

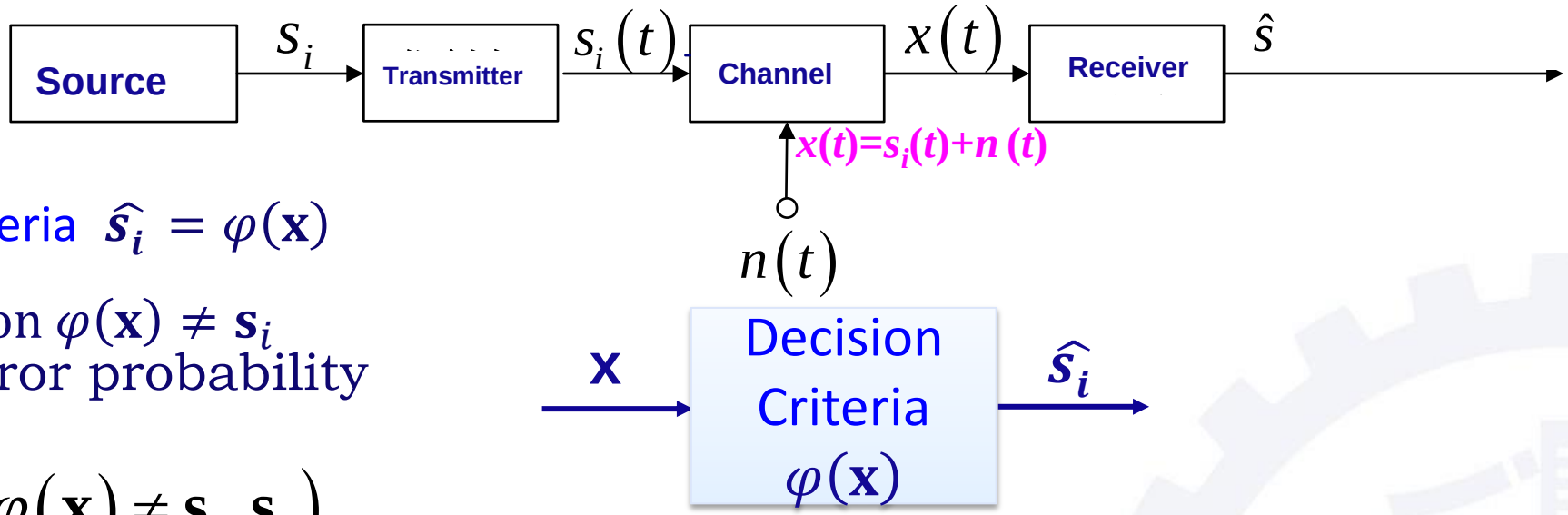
RECEIVED SIGNAL VECTOR OF DIGITAL COMMUNICATION SYSTEM: SUMMARY

- Noise's vector components falling outside the signal space are independent of the signal
- Distance between received signal and sample signal is only measured in signal space

- Digital signal optimum receiver
- MAP criteria
- Optimum receiver

7.2 OPTIMUM RECEIVER PROBLEM AND MAP CRITERIA

Optimum receiver problem



- Receiver
  - Decision Criteria $\widehat{s_i} = \varphi(\mathbf{x})$
  - Transmit $\mathbf{s}_i$, error decision $\varphi(\mathbf{x}) \neq \mathbf{s}_i$
  - Average error probability

$$P_e = \sum_{i=1}^{M} P\left(\varphi(\mathbf{x}) \neq \mathbf{s}_i, \mathbf{s}_i\right)$$

- Optimum receiver
  - Given noise conditions, receiver with smallest P_e
  - How to design decision criteria?
  - How to implement the decision criterion?

Several probabilities

- *a priori* probability
  - Priori probability of transmitting $s_1(t)$ or $s_2(t)$ without any observation

$$P\left(s_1\left(t\right)\right)=P_1 \qquad P\left(s_2\left(t\right)\right)=P_2$$

- *a posteriori* probability
  - probability of transmitting $s_1(t)$ or $s_2(t)$ under condition that received signal $x(t)$ is observed

$$P\left(s_1\left(t\right)\middle|x\left(t\right)\right) \qquad P\left(s_2\left(t\right)\middle|x\left(t\right)\right)$$

- likelihood probability
  - Under condition of transmitting $s_1(t)$ or $s_2(t)$, probability of observing received signal $x(t)$ at receiver
  - Channel transition probability

$$P\left(x\left(t\right)\middle|s_1\left(t\right)\right) \quad P\left(x\left(t\right)\middle|s_2\left(t\right)\right)$$

MAP criterion and optimum receiver model

- MAP criterion — Maximum *a posteriori* probability

$$\hat{s} = \arg\max_{1 \le i \le M} P\left(s_i(t) \middle| x(t)\right)$$

  - Ensure maximum probability of correct decision, that is, minimum bit error rate

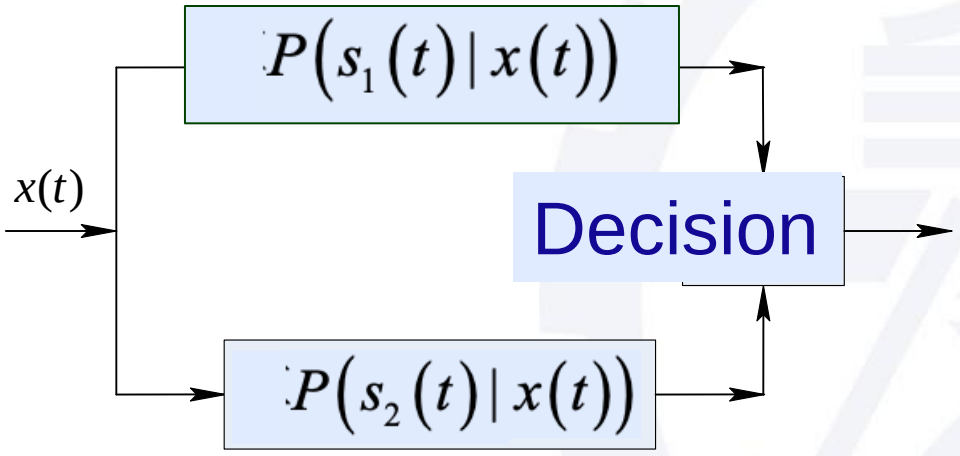
- optimum receiver model
  - How to use electronic devices to calculate posterior probability?
  - Unrealizable: : $x(t)$ is a random signal

How to make optimum receiver possible?

Convert conditional probability in form of a random function to conditional probability in form of a random variable

Binary case

$$P\left(s_1(t) \middle| x(t)\right) \underset{s_2(t)}{\overset{s_1(t)}{\gtrless}} P\left(s_2(t) \middle| x(t)\right)$$



Optimum decision criterion vs Optimum receiver

Optimum decision criterion	Optimum receiver
Maximum posterior probability criterion MAP	Optimum receiver
Maximum likelihood criterion ML	Correlation receiver
Maximum output signal-to-noise ratio criterion	Matched filter method

Equivalent conditions of Optimum receiver criteria?
Optimum receiver design principle?

OPTIMUM RECEIVER PROBLEMS AND MAP CRITERION: SUMMARY

- Optimum receiver minimizes bit error rate
- From perspective of receiver, maximum posterior probability (probability of correct decision) means minimum error rate
- Three optimum receiver criteria vs optimum receiver

ML criterion and signal distance

Correlation receivers

7.3 ML CRITERIA AND CORRELATION RECEIVERS

Maximum likelihood criterion

- Maximum posterior criterion

$$\hat{s} = \arg\max_{1 \le i \le M} P\left(s_i(t) \middle| x(t)\right)$$

$$P\left(s_i(t) \middle| x(t)\right) = \underbrace{P\left(x(t) \middle| s_i(t)\right)}_{\text{Likelihood}} \cdot \frac{P\left(s_i(t)\right)}{P\left(x(t)\right)}$$

Has nothing to do with i

Convert posterior probability calculation to likelihood probability calculation

$$\hat{s} = \arg\max_{1 \le i \le M} P\left(s_i(t)\right) P\left(x(t) \middle| s_i(t)\right)$$

M=2

$$\frac{P\left(x(t) \middle| s_1(t)\right)}{P\left(x(t) \middle| s_2(t)\right)} \underset{s_2}{\overset{s_1}{\gtrless}} \frac{P\left(s_2(t)\right)}{P\left(s_1(t)\right)}$$

Maximum likelihood ratio criterion

a priori equivalent

Maximum Likelihood， ML

$$\hat{s} = \arg\max_{1 \le i \le M} P\left(x(t) \middle| s_i(t)\right)$$

M=2

$$P\left(x(t) \middle| s_1(t)\right) \underset{s_2}{\overset{s_1}{\gtrless}} P\left(x(t) \middle| s_2(t)\right)$$

equivalent to MAP guidelines?

Likelihood function

$$x_i = \int_{-\infty}^{\infty} x(t) \cdot f_i(t) dt$$

AWGN channel

$$= \int_{-\infty}^{\infty} \left[s(t) + n(t) \right] f_i(t) dt = s_i + n_i$$

- $x_1 = s_{11} + n_1$, $x_2 = s_{12} + n_2$
- Independent Gaussian processes

$$p\left(x_1 \middle| s_{i1}\right) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{\left(x_1 - s_{i1}\right)^2}{2\sigma^2}}$$

$$p\left(x_2 \middle| s_{i2}\right) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{\left(x_2 - s_{i2}\right)^2}{2\sigma^2}}$$

$$P\left(x(t) \middle| s_i(t)\right)$$

$$= P\left(\mathbf{x} \middle| \mathbf{s_i}\right)$$

$$= P\left(x_1, x_2 \middle| s_{i1}, s_{i2}\right)$$

$$= P\left(x_1 \middle| s_{i1}\right) P\left(x_2 \middle| s_{i2}\right)$$

$$= \frac{1}{2\pi\sigma^2} e^{-\frac{\left(x_1 - s_{i1}\right)^2 + \left(x_2 - s_{i2}\right)^2}{2\sigma^2}}$$

$$d\left[\mathbf{x}, \mathbf{s}_1\right] = \sqrt{\left(x_1 - s_{11}\right)^2 + \left(x_2 - s_{12}\right)^2}$$

$$= \sqrt{\int_{-\infty}^{\infty} \left(x(t) - s_1(t)\right)^2 dt - E_v}$$

distance between received signal and sample signal in signal space

Maximum likelihood function
Equivalent to minimum signal distance

Realization of maximum likelihood criterion

AWGN Channel

ML criterion

$$P\left(x(t)\middle|s_1(t)\right) \underset{s_2}{\overset{s_1}{\gtrless}} P\left(x(t)\middle|s_2(t)\right)$$

$$p\left(x_1, x_2\middle| s_{i1}, s_{i2}\right) = \frac{1}{2\pi\sigma^2} e^{-\frac{\left(x_1-s_{i1}\right)^2+\left(x_2-s_{i2}\right)^2}{2\sigma^2}}$$

Minimum distance criterion

$$\int_{-\infty}^{\infty}\left(x(t)-s_2(t)\right)^2 dt \underset{s_2}{\overset{s_1}{\gtrless}} \int_{-\infty}^{\infty}\left(x(t)-s_1(t)\right)^2 dt$$

Tidy up

Signal distance calculation can be simplified to correlation function calculation

$$\int_{-\infty}^{\infty} x(t)s_1(t)dt - \frac{E_1}{2} \underset{s_2}{\overset{s_1}{\gtrless}} \int_{-\infty}^{\infty} x(t)s_2(t)dt - \frac{E_2}{2}$$

Correlators

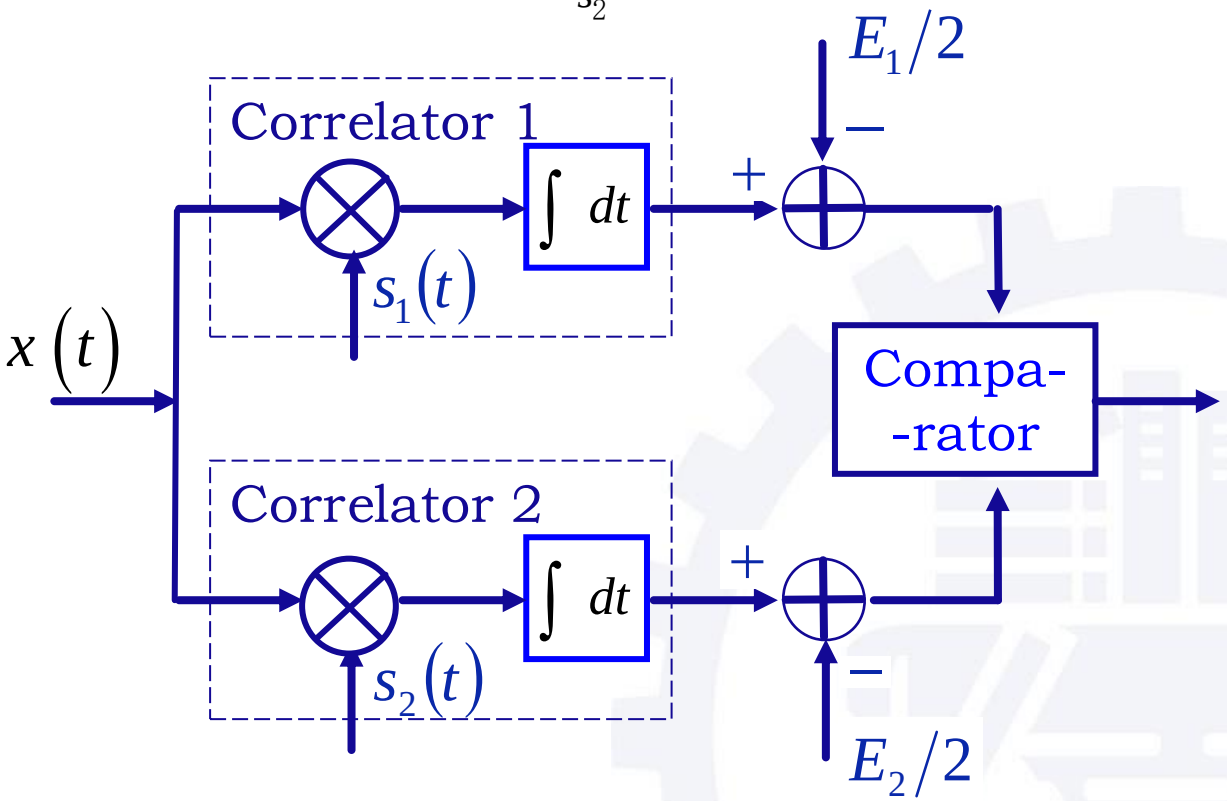
Correlators

Correlation receivers

Correlation receivers

- Adopt correlator to realize cross correlation
- Correlator for each sample signal
- Comparator decision
  - Select sample signal with greatest correlation with received signal as estimated value

$$\int_{-\infty}^{\infty} x(t)s_1(t)dt - \frac{E_1}{2} \underset{s_2}{\overset{s_1}{\gtrless}} \int_{-\infty}^{\infty} x(t)s_2(t)dt - \frac{E_2}{2}$$



If $E_1 = E_2$ compensation term can be removed

Correlator output signal

Correlator 1 output signal

$$\int_{-\infty}^{\infty} x(t)s_1(t)dt \underset{s_2}{\overset{s_1}{\gtrless}} \int_{-\infty}^{\infty} x(t)s_2(t)dt$$

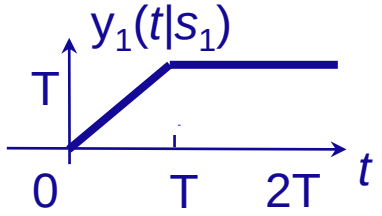
Ignore channel noise,

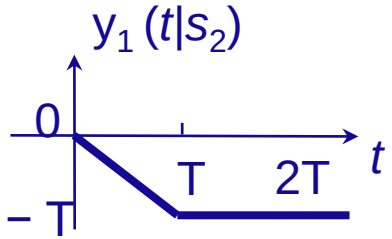
Transmit symbol 1

$$y_1(t\,|s_1) = \int_{-\infty}^{t} s_1(t)s_1(t)dt = \begin{cases} t, & 0 \le t \le T \\ T, & t > T \end{cases}$$

Transmit symbol 0

$$y_1(t\,|s_2) = \int_{-\infty}^{t} s_2(t)s_1(t)dt = \begin{cases} -t, & 0 \le t \le T \\ -T, & t > T \end{cases}$$

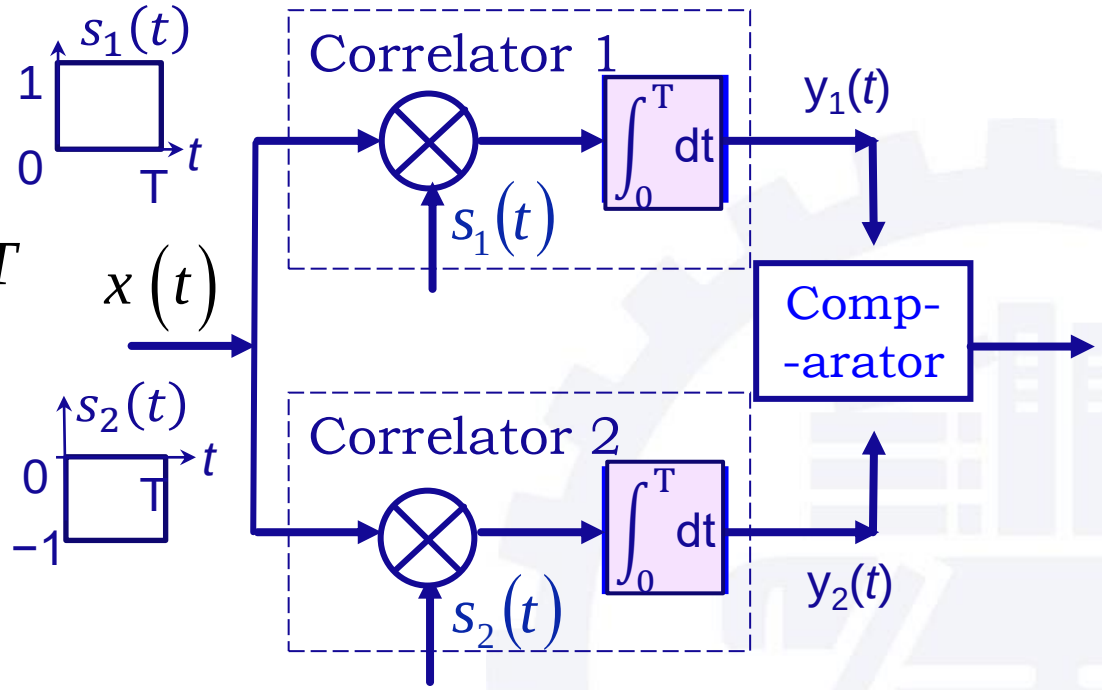




Cleared after each symbol period
Avoid accumulated affecting on subsequent symbols

Bipolar code

$E_1 = E_2$



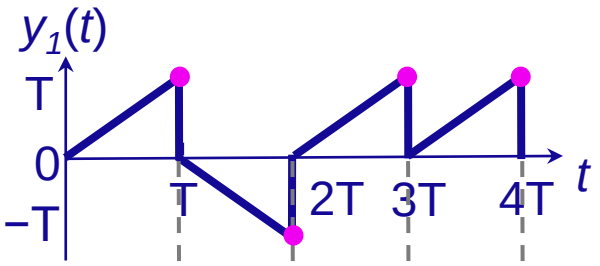
Correlator output signal

$$\int_{-\infty}^{\infty} x(t)s_1(t)dt \underset{s_2}{\overset{s_1}{\gtrless}} \int_{-\infty}^{\infty} x(t)s_2(t)dt$$

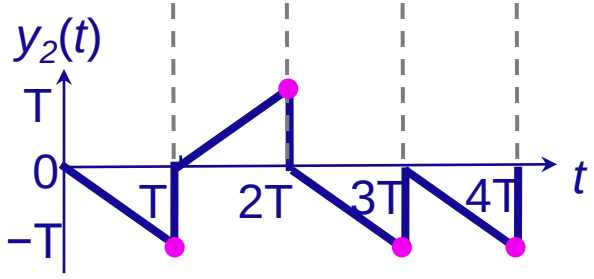
Ignore channel noise

Input sequence 1 0 1 1

Correlator 1 output signal



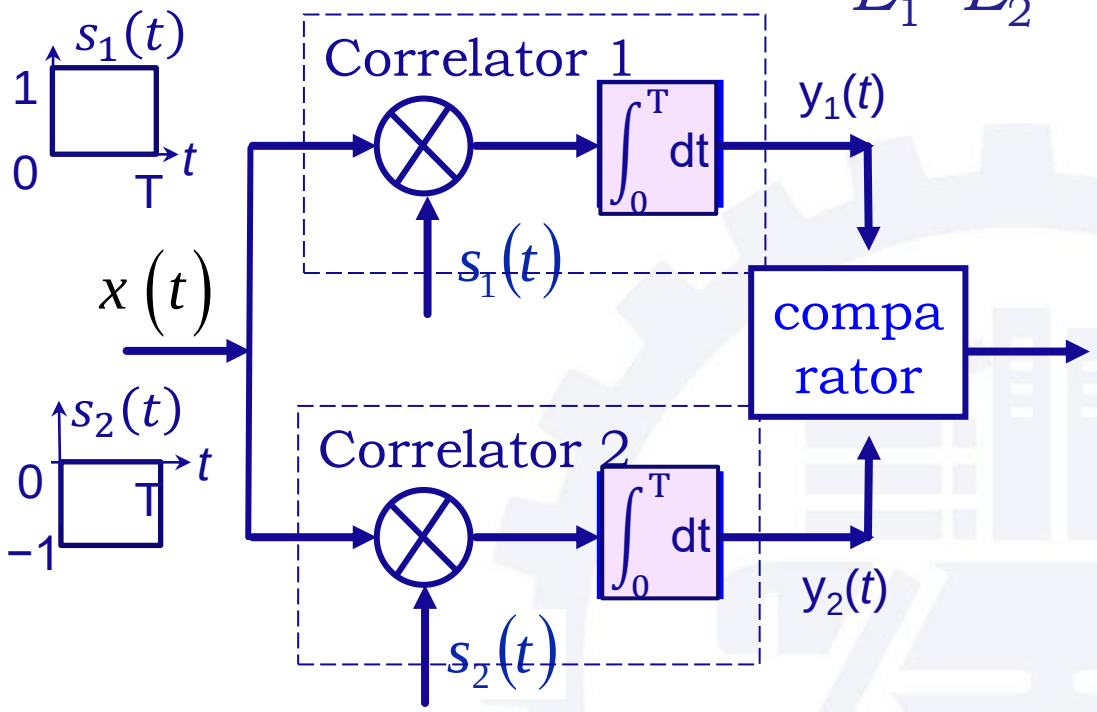
Correlator 2 output signal



1 0 1 1

Bipolar code

$E_1 = E_2$



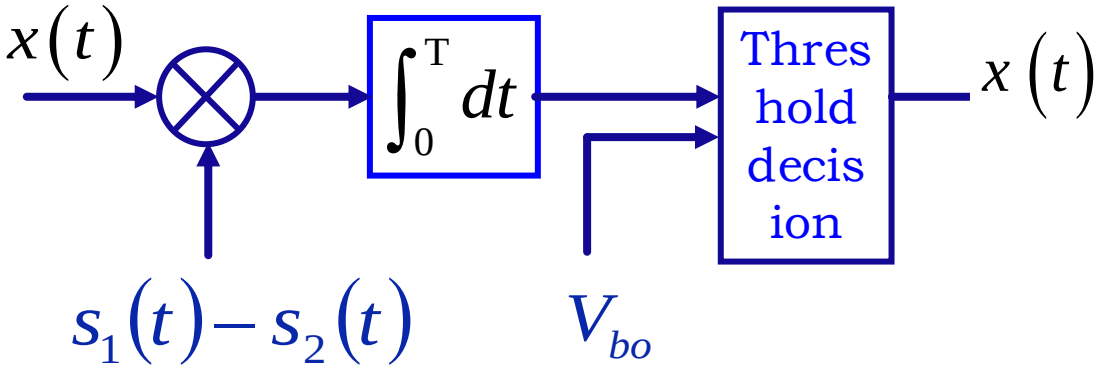
Sampling decision

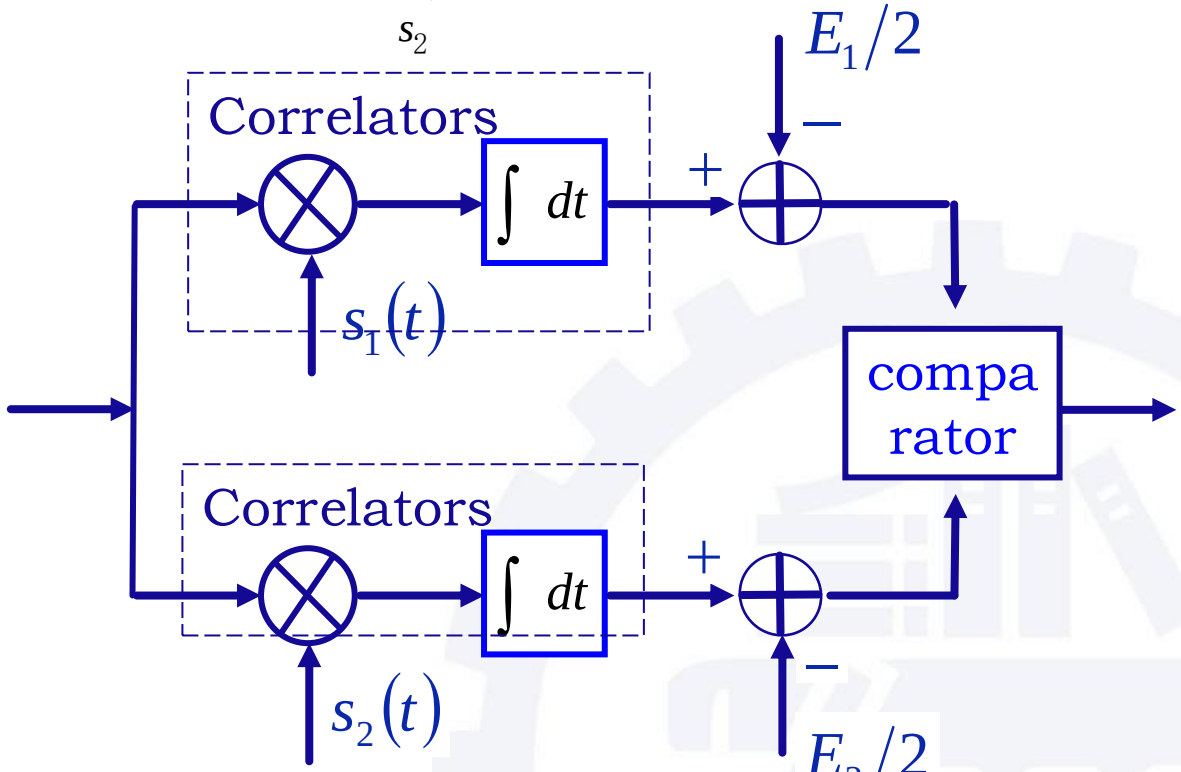
Simplified implementation

$$\int_{-\infty}^{\infty} x(t)s_1(t)dt - \frac{E_1}{2} \underset{s_2}{\overset{s_1}{\gtrless}} \int_{-\infty}^{\infty} x(t)s_2(t)dt - \frac{E_2}{2}$$

- Use only 1 correlator
- Threshold decision

$$\int_{-\infty}^{\infty} x(t)\left(s_1(t)-s_2(t)\right)dt \underset{s_2}{\overset{s_1}{\gtrless}} V_{bo} = \frac{E_1 - E_2}{2}$$





ML CRITERIA AND CORRELATION RECEIVERS: SUMMARY

- Under equal prior probability, ML criterion is equivalent to MAP criterion
- In AWGN channel, ML criterion is equivalent to minimum signal distance criterion
- Distance calculation can be converted into correlation calculations to achieve correlation receiver

- Matched filter design principles
- Optimum receiver using matched filter

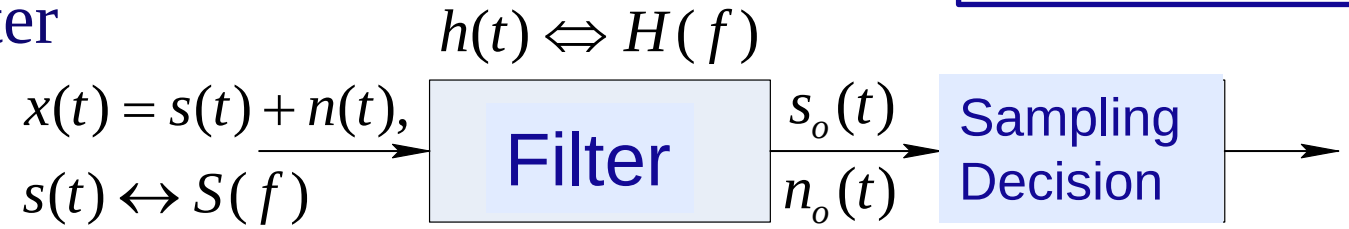
7.4 MATCHED FILTER OPTIMUM RECEIVER

7.4.1 DESIGN PRINCIPLE OF MATCHED FILTER

Maximum output signal-to-noise ratio criterion

AWGN channel

- Treat receiver as a filter
- Filter output



$$s_o(t) = s(t) * h(t)$$

$$= \int_{-\infty}^{\infty} H(f) S(f) e^{j2\pi ft} df$$

$$N_o = \overline{n_o^2(t)} = \int_{-\infty}^{\infty} \frac{n_0}{2} |H(f)|^2 \, df$$

- Instantaneous output signal-to-noise ratio

$$\gamma_0 = \frac{|s_o(t_0)|^2}{N_o} = \frac{\left| \int_{-\infty}^{\infty} H(f) S(f) e^{j2\pi ft_0} df \right|^2}{\frac{n_0}{2} \int_{-\infty}^{\infty} |H(f)|^2 \, df}$$

Select filter H (f), Ensure maximum instantaneous output signal-to-noise ratio at the sample time

Matched filter transfer function

Schwartz inequality

$$\left|\int_{-\infty}^{\infty} X(f)Y(f)df\right|^2 \le \int_{-\infty}^{\infty}|X(f)|^2 df \cdot \int_{-\infty}^{\infty}|Y(f)|^2 df$$

Equal conditions: $X(f) = k \cdot Y^*(f)$

- Instantaneous output signal-to-noise ratio

$$\gamma_o = \frac{|s_o(t_0)|^2}{N} = \frac{2}{n_0} \cdot \frac{\left|\int_{-\infty}^{\infty} H(f)S(f)e^{j2\pi f t_0} df\right|^2}{\int_{-\infty}^{\infty}|H(f)|^2 df}$$

$$E = \int_{-\infty}^{\infty}|S(f)|^2 df = \int_0^T s^2(t)dt$$

- Maximum output signal to noise ratio

$$\gamma_{o,\max} = \frac{2}{n_0} \cdot \frac{\int_{-\infty}^{\infty}|H(f)|^2 df \cdot \int_{-\infty}^{\infty}|S(f)|^2 df}{\int_{-\infty}^{\infty}|H(f)|^2 df}$$

$$= \frac{2}{n_0} \cdot \int_{-\infty}^{\infty}|S(f)|^2 df = \frac{2E}{n_0}$$

Ratio of signal energy ***E*** **to** noise power spectral density n_0, unrelated to signal waveform

- Transfer Function $H(f) = k \cdot S^*(f)e^{-j2\pi f t_0}$ Usually k=1

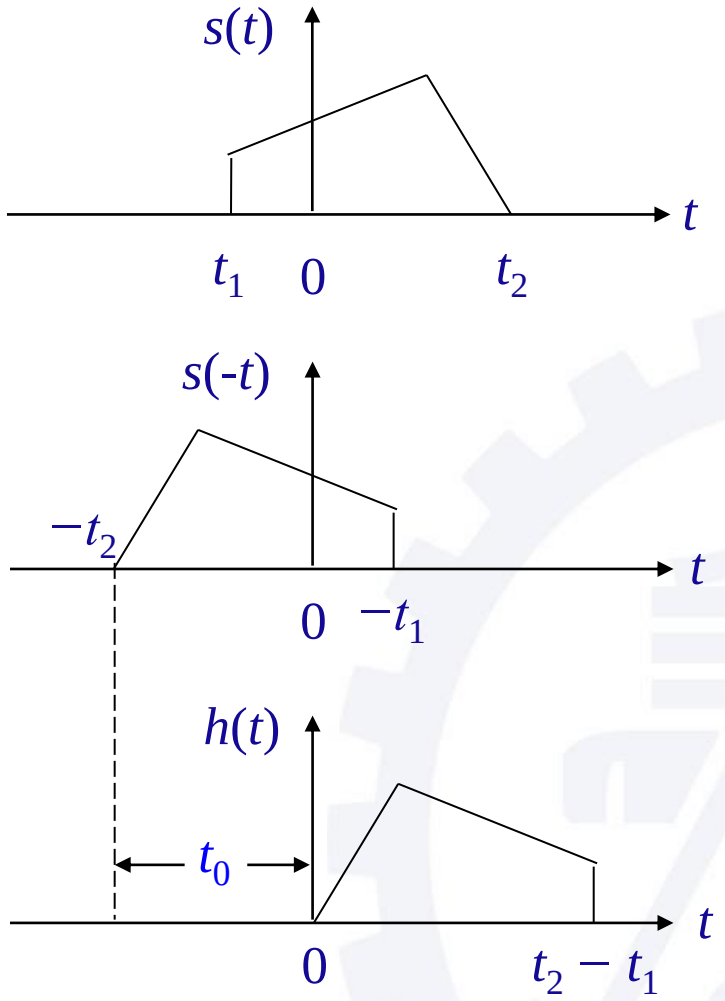
MF: Matched filter

Complex conjugate of signal symbol spectrum

Matched filter impulse response

$$H(f) = k \cdot S^*(f) e^{-j2\pi f t_0}$$

$$\begin{aligned}
h(t) &= \int_{-\infty}^{\infty} H(f) e^{j2\pi ft} df \\
&= \int_{-\infty}^{\infty} kS^*(f) e^{-j2\pi ft_0} e^{j2\pi ft} df \\
&= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} ks(\tau) e^{-j2\pi f\tau} d\tau \right]^* e^{-j2\pi f(t_0 - t)} df \\
&= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} e^{j2\pi f(\tau - t_0 + t)} df \right] ks(\tau) d\tau \\
&= \int_{-\infty}^{\infty} ks(\tau) \delta(\tau - t_0 + t) d\tau \\
&= ks(t_0 - t) \qquad \text{Usually } k=1
\end{aligned}$$



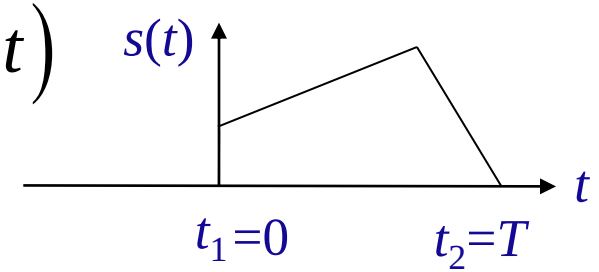
- $s(t)$ mirror image $s(-t)$, shifted right by t_0 on time axis

Sampling time t_0

$$h(t) = \int_{-\infty}^{\infty} H(f)e^{j2\pi ft}df$$

$$= s\left(t_0 - t\right)$$

$$= s(T - t)$$



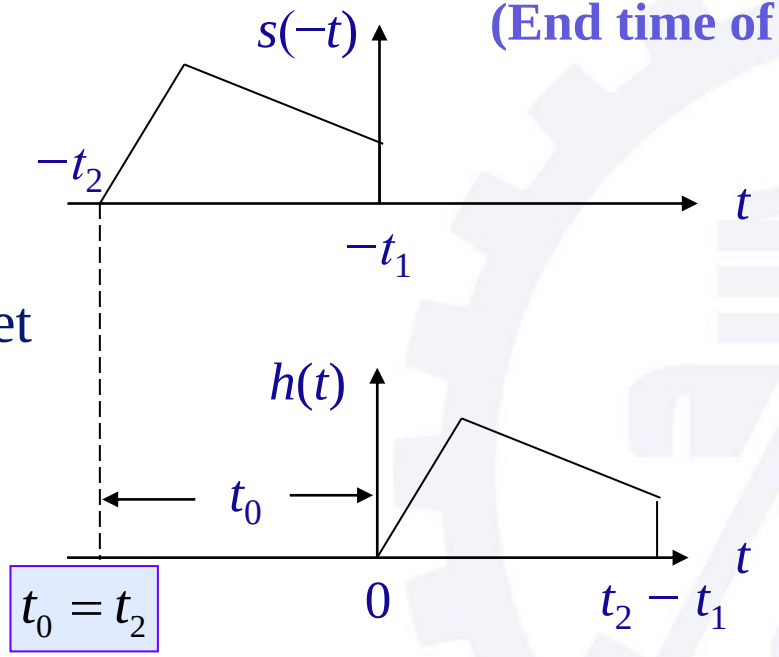
- In impulse response function t_0=?

- Filter physical realizability
  - When $t<0$, $h(t)=0$, i.e., $s(t_0-t)=0$
  - So, when $t>t_0$, $s(t)=0$

$s(t)$ rightmost coordinate

(End time of symbol signal)

- t_0: moment when $s(t)$ disappears: $t_0 \geq t_2$
  - t_0 physical meaning: sampling moment at receiver
  - Sampling after signal is completely input, to get all information
  - larger t_0 means longer reception delay, so take equal sign
  - Usually $t_1 = 0$, $t_2 = \mathrm{T}$, so $t_0 = T$



Matched filter output

- Output signal

$$s_o(t) = s(t) * h(t) = \int_{-\infty}^{\infty} s(t-\tau)h(\tau)d\tau$$

$$= \int_{-\infty}^{\infty} s(t-\tau)s(t_0-\tau)d\tau$$

$$= R(t-t_0) \sim \text{Correlation}$$

- Output noise

$$N_o = \overline{n_o^2(t)} = \int_{-\infty}^{\infty} \frac{n_0}{2}\left|H(f)\right|^2 df$$

$$= \int_{-\infty}^{\infty} \frac{n_0}{2}\left|S^*(f)e^{-j2\pi ft_0}\right|^2 df = \frac{n_0 E}{2}$$

- Output signal sampling value

$$s_o(t_0) = R(0) = E$$

- Instantaneous output signal-to-noise ratio

$$\gamma_0 = \frac{\left|s_o(t_0)\right|^2}{N_o} = \frac{2E}{n_0}$$

$$= \gamma_{o,\max}$$

Output signal sampling value is signal energy value, screening phase effects

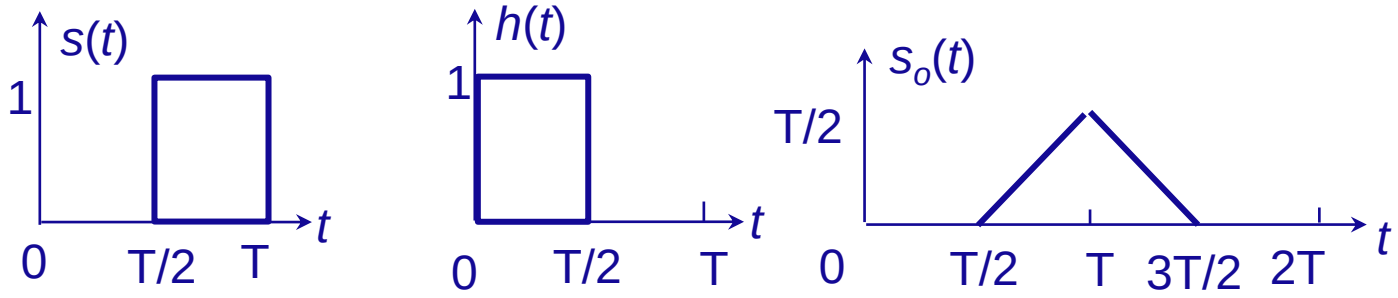
Example

Given white noise of input noise bilateral power spectral density of $n_0/2$, signal $s(t)$ is shown in figure, design its matched filter

(1) Draw impulse response waveform of matched filter $h(t)$

(2) Draw matched output waveform of matched filter $s_o(t)$

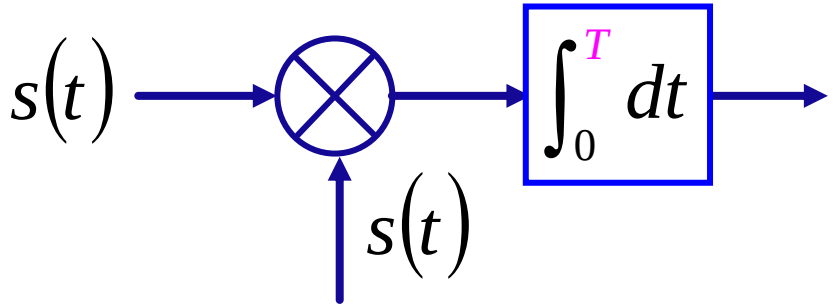
(3) Calculate maximum output signal-to-noise ratio of matched filter



$$E = \int_0^T s^2(t)dt = \frac{T}{2} \qquad \gamma_{o,\max} = \frac{2E}{n_0} = \frac{T}{n_0}$$

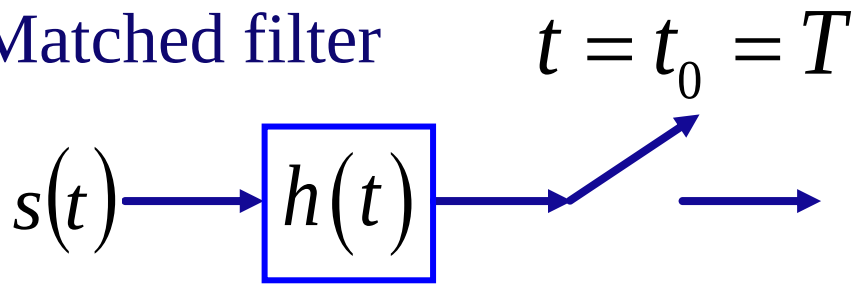
Equivalent relationship between matched filter and correlator

- Correlators



$$y_1 = \int_0^T s(t)\, s(t)\, dt = R(0)$$

- Matched filter



$$y_2(t) = \int_{-\infty}^{\infty} s(\tau)\, h(t-\tau)\, d\tau = R(t-T)$$

$$y_2(T) = R(0)$$

Same decision sampling value

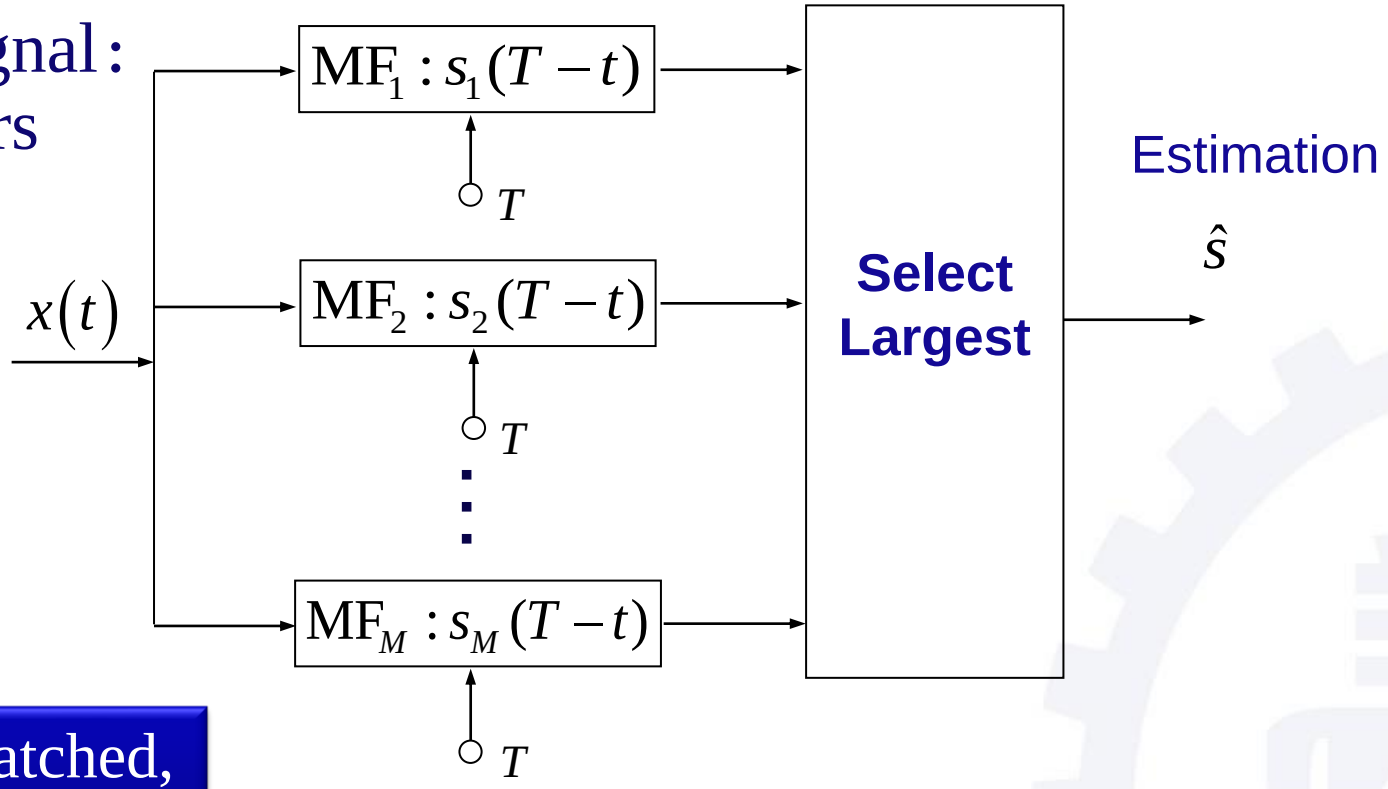
MATCHED FILTER DESIGN PRINCIPLES SUMMARY

- Matched filter has largest instantaneous output signal-to-noise ratio
- Maximum instantaneous output signal-to-noise ratio is determined by signal energy and noise power spectral density
- Matched filter output is delayed input signal autocorrelation function

7.4.2 MATCHED FILTER RECEIVER

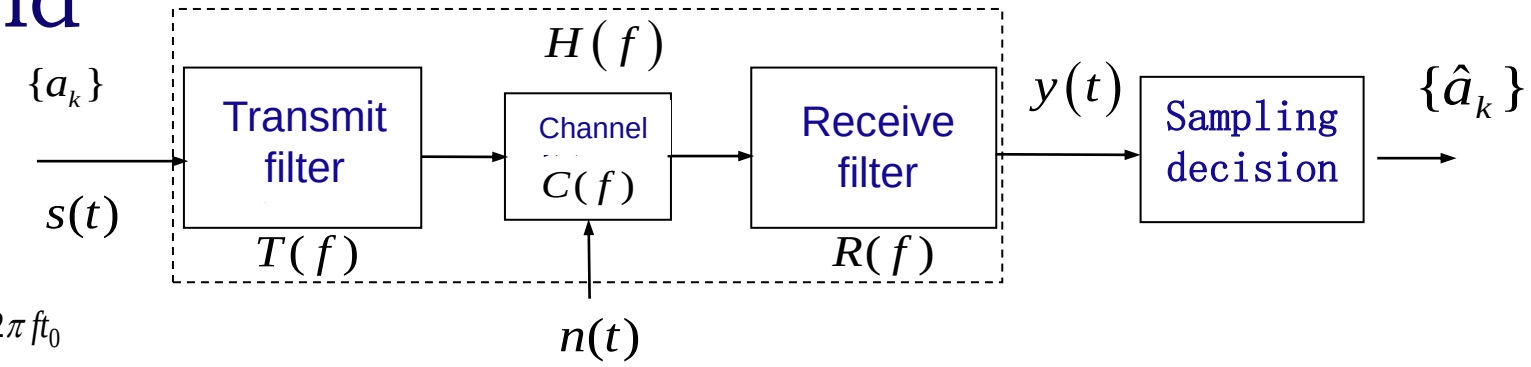
Matched filtering optimum receiver

- M-ary digital signal : M matched filters



If base signal is matched, how to design receiver?

Optimum of baseband transmission



System transfer function

$$H(f)=T(f)\cdot C(f)\cdot R(f)=|H(f)|e^{-j2\pi ft_0}$$

$$=T(f)\cdot R(f)=|H(f)|e^{-j2\pi f(t_T+t_R)}$$

To achieve optimal receiver, receive filter should match transmit filter

$$T(f)=\sqrt{|H(f)|}e^{-j2\pi ft_T}$$

$$R(f)=\sqrt{|H(f)|}e^{-j2\pi ft_R}$$

If a raised cosine pulse is used, both transmit filter and receive filter are root raised cosine filters

Assume perfectly equalized channel

$$C(f)=e^{-j2\pi ft_C}$$

Amplitude-frequency characteristic

$$|H(f)|=|T(f)|\cdot|C(f)|\cdot|R(f)|$$

Phase frequency characteristics

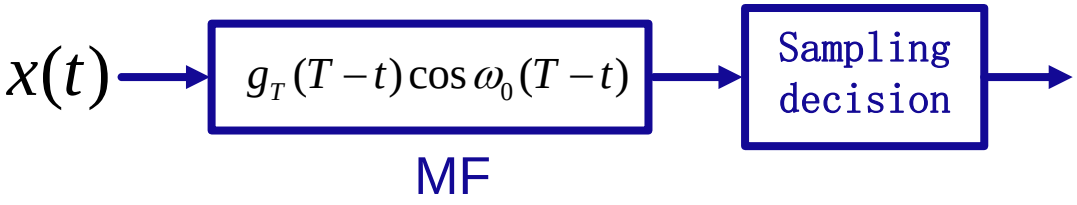
$$\phi_T(f)+\phi_C(f)+\phi_R(f)=-j2\pi ft_0$$

Total system delay

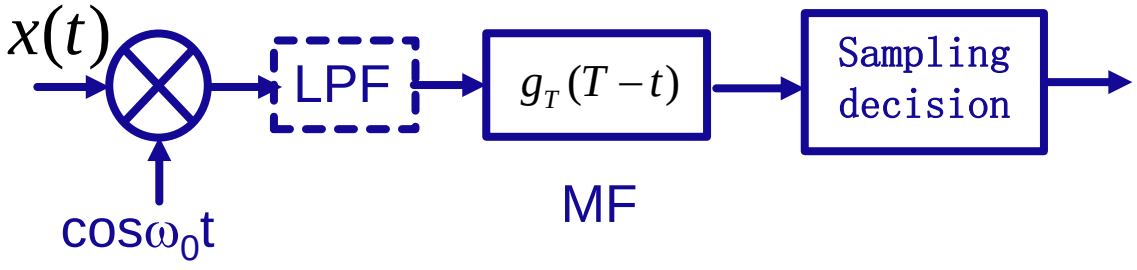
$$t_0=t_T+t_C+t_R$$

Equivalent form of optimum receiver of frequency band signals

- Frequency band detection



- Coherent demodulation + baseband detection



Match detection in baseband, looser requirements for synchronization timing

MATCHED FILTER OPTIMUM RECEIVER SUMMARY

- Receiver filter and transmitting filter of baseband transmission system are matched to achieve optimum receiver
- After demodulation of digital frequency band signal, requirement for synchronization can be relaxed in baseband matching detection

- Bit error rate for optimal receiver of binary signals
- Comparison of bit error rate for baseband signal optimum receiver
- Comparison of bit error rate for frequency band signal optimum receiver

6.5 ERROR PERFORMANCE ANALYSIS OF OPTIMUM RECEIVER

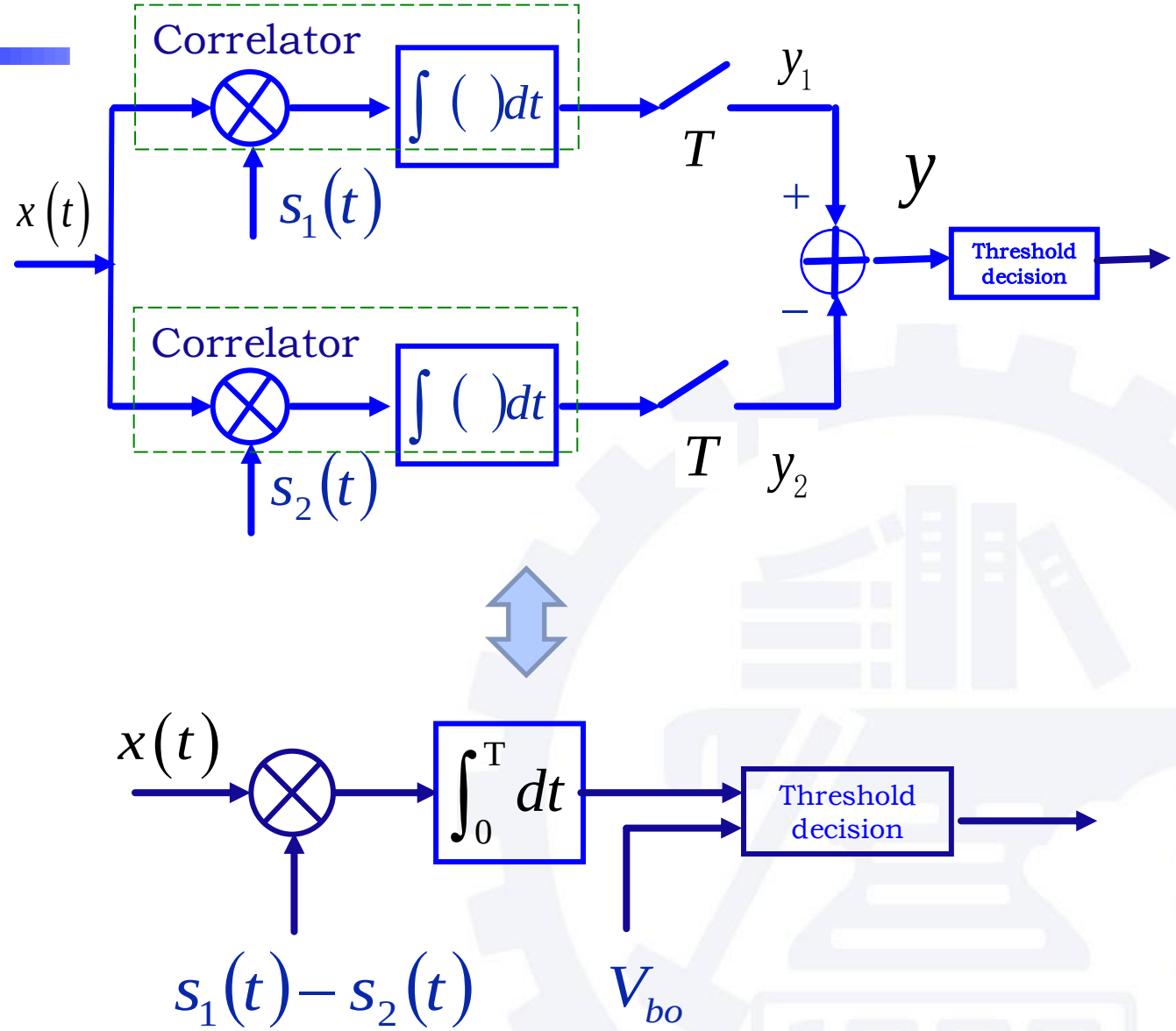
Binary signal optimum receiver

- Correlation receiver as example, Case of equal probability
- Average bit error rate

$$P_e = P(s_1)P(s_2 \mid s_1) + P(s_2)P(s_1 \mid s_2)$$

$$\int_0^T x(t)s_1(t)dt - \int_0^T x(t)s_2(t)dt \;\underset{s_2(t)}{\overset{s_1(t)}{\gtrless}}\; V_{bo}$$

$$\underbrace{\int_0^T x(t)\left[s_1(t) - s_2(t)\right]dt}_{y} \;\underset{s_2(t)}{\overset{s_1(t)}{\gtrless}}\; V_{bo}$$



Error transfer probability

Transmit s_1

$$x(t) = s_1(t) + n(t)$$

Misjudged as s_2

$$P(s_2 \mid s_1) = P(y < V_{bo})$$

$$y = \int_0^T x(t)\left[s_1(t) - s_2(t)\right]dt$$

Correlator output

$$= \int_0^T \left[s_1(t) + n(t)\right]\left[s_1(t) - s_2(t)\right]dt$$

$$= \int_0^T s_1^2(t)dt - \int_0^T s_1(t)s_2(t)dt + \int_0^T \left[s_1(t) - s_2(t)\right]n(t)dt$$

$$= E_1 - E_{12} + \int_0^T \left[s_1(t) - s_2(t)\right]n(t)dt$$

Energy of $s_1(t)$

Mutual energy of signals $s_1(t)$ and $s_2(t)$

Sample signal distance

Stationary Gaussian process

Decision rule

$$\frac{\int_0^T x(t)\left[s_1(t) - s_2(t)\right]dt}{y} \begin{array}{c} s_1(t) \\ > \\ < \\ s_2(t) \end{array} V_{bo}$$

y: Stationary Gaussian random variable

Mean? variance?

Statistical characteristics of random variable y

$$y = E_1 - E_{12} + \int_0^T \left[s_1(t) - s_2(t) \right] n(t) dt$$

$$E[y] = E_1 - E_{12} + E\left[\int_0^{T_s} \left[s_1(t) - s_2(t) \right] n(t) dt \right] = E_1 - E_{12} = m_y$$

$\int_0^{T_s} \left[s_1(t) - s_2(t) \right] n(t) dt \to 0$

Choose different time t, u
Get two random variables

$$D[y] = E\left[\left(y - m_y \right)^2 \right] = E\left[\int_0^T \left[s_1(t) - s_2(t) \right] n(t) dt \int_0^T \left[s_1(u) - s_2(u) \right] n(u) du \right]$$

Order Exchange of integral and expectation

$$= \int_0^T \left[\int_0^T \left[s_1(t) - s_2(t) \right] \left[s_1(u) - s_2(u) \right] E\left[n(t) n(u) \right] du \right] dt$$

Gaussian white noise autocorrelation function

$$R_n(t-u) = E\left[n(t) n(u) \right] = \frac{n_0}{2} \delta(t-u)$$

$$= \int_0^T \frac{n_0}{2} \left[s_1(u) - s_2(u) \right] \int_0^T \left[s_1(t) - s_2(t) \right] \delta(t-u) dt du$$

$$= \frac{n_0}{2} \int_0^T \left[s_1(u) - s_2(u) \right]^2 du$$

Sampling with impulse signals

$$\int_0^T \delta(t-u) f(t) dt = f(u)$$

$$= \frac{n_0}{2} \left(E_1 + E_2 - 2E_{12} \right) = \sigma_y^2$$

$$p\left(y \mid s_1 \right) \sim N\left(E_1 - E_{12}, \frac{n_0}{2} \left(E_1 + E_2 - 2E_{12} \right) \right)$$

Transmit s_1, probability density function of the correlator output y

Bit error rate

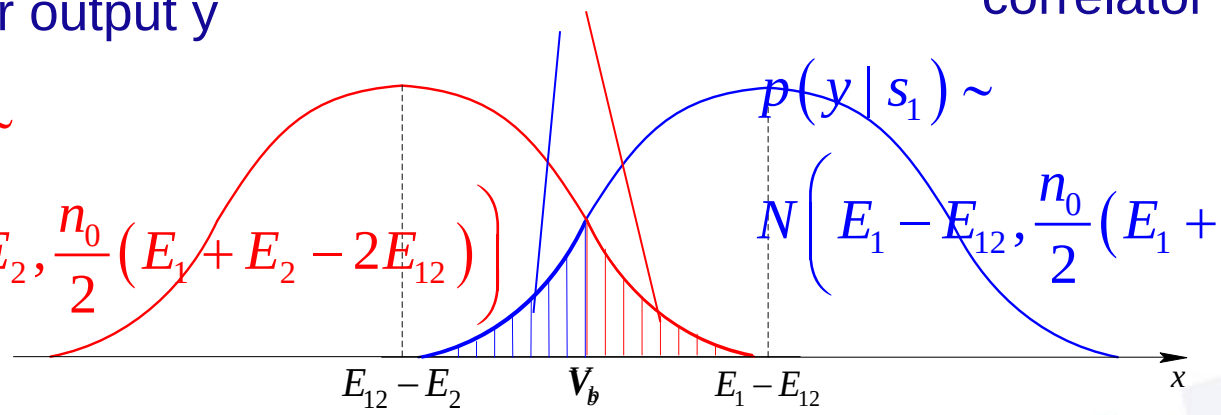
Transmit s_2, Probability density function of correlator output y

Error transfer probability

Transmit s_1, Probability density function of correlator output y

$$p(y \mid s_2) \sim N\left(E_{12}-E_2, \frac{n_0}{2}\left(E_1+E_2-2E_{12}\right)\right)$$

$$p(y \mid s_1) \sim N\left(E_1-E_{12}, \frac{n_0}{2}\left(E_1+E_2-2E_{12}\right)\right)$$



Probability density function of correlator output y

Equal prior probability

$$V_{bo} = \frac{E_1+E_2-2E_{12}}{2}$$

$$P_e = P(s_1)P(s_2 \mid s_1) + P(s_2)P(s_1 \mid s_2)$$

$$= \frac{1}{2}\operatorname{erfc}\left(\sqrt{\frac{E_1+E_2-2E_{12}}{4n_0}}\right)$$

$$= \frac{1}{2}erfc\left(\sqrt{\frac{E_b(1-\rho)}{2n_0}}\right)$$

$$\rho = \frac{1}{\sqrt{E_1E_2}}\int_{-\infty}^{\infty} s_1(t)s_2(t)dt = \frac{E_{12}}{\sqrt{E_1E_2}}$$

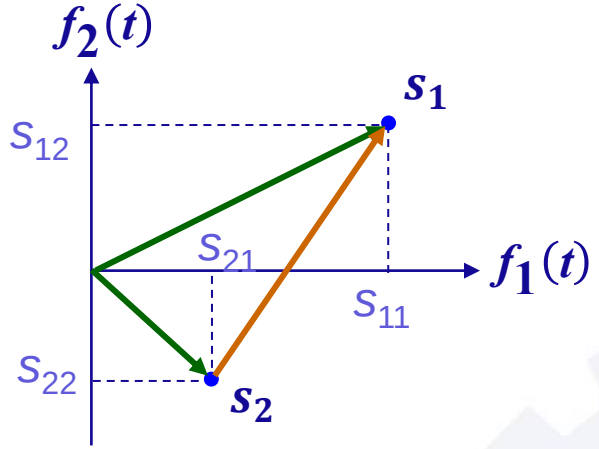
$$E_b = \frac{E_1+E_2}{2}$$ Average bit energy

E_b/n_0 Equivalent to signal to noise power ratio

Bit error rate and constellation point distance

$$P_e = \frac{1}{2}\operatorname{erfc}\left(\sqrt{\frac{E_b(1-\rho)}{2n_0}}\right)$$

$$= \frac{1}{2}erfc\left(\sqrt{\frac{d^2}{4n_0}}\right)$$



Euclidean distance between sample signals

Only correlated to constellation point distance and channel noise

$$d_{12} = \|\mathbf{s}_1 - \mathbf{s}_2\| = \sqrt{\sum_i (s_{1i} - s_{1i})^2}$$

$$= \sqrt{E_1 + E_2 - 2\sqrt{E_1 E_2}\,\rho}$$

$$= \sqrt{2E_b(1-\rho)}$$

Conclusion applies to matched filtering optimum receiver

Binary baseband signal optimum receiver bit error rate

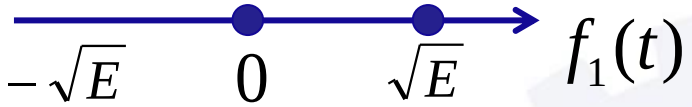
$$P_e = \frac{1}{2} erfc\left(\sqrt{\frac{E_b(1-\rho)}{4n_0}}\right) = \frac{1}{2} erfc\left(\sqrt{\frac{d^2}{4n_0}}\right)$$

- Unipolar baseband signal

$$E_1 = A^2 T_b, E_2 = 0, \rho_{12} = 0, d = \sqrt{E_1} = 2\sqrt{E_b}$$

$$E_b = \frac{E_1 + E_2}{2} = \frac{E_1}{2}$$

$$P_e = \frac{1}{2} erfc\left(\sqrt{\frac{E_b}{2n_0}}\right) = \frac{1}{2} erfc\left(\sqrt{\frac{E_1}{4n_0}}\right)$$



- Bipolar baseband signal

$$E_1 = E_2 = E_b = A^2 T_b, \rho_{12} = -1, d = 2\sqrt{E_b}$$

$$P_e = \frac{1}{2} erfc\left(\sqrt{\frac{E_b}{n_0}}\right)$$

E_1 for unipolar signals is the same as E_b for bipolar signals

At same bit error rate, E_b/n_0 of bipolar signals can save 3dB

Binary frequency band signal optimum receiver bit error rate

$$P_e = \frac{1}{2}erfc\left(\sqrt{\frac{E_b(1-\rho)}{4n_0}}\right) = \frac{1}{2}erfc\left(\sqrt{\frac{d^2}{4n_0}}\right)$$

- ASK signal

$E_1 = A^2T_b/2,\ \ E_2 = 0,$

$\rho_{12} = 0,\ \ d = \sqrt{E_1} = 2\sqrt{E_b}$

$$P_e = \frac{1}{2}erfc\left(\sqrt{\frac{E_1}{4n_0}}\right)$$

- PSK signal

$E_1 = E_2 = E_b = A^2T_b/2,$

$\rho_{12} = -1,\ \ d = 2\sqrt{E_b}$

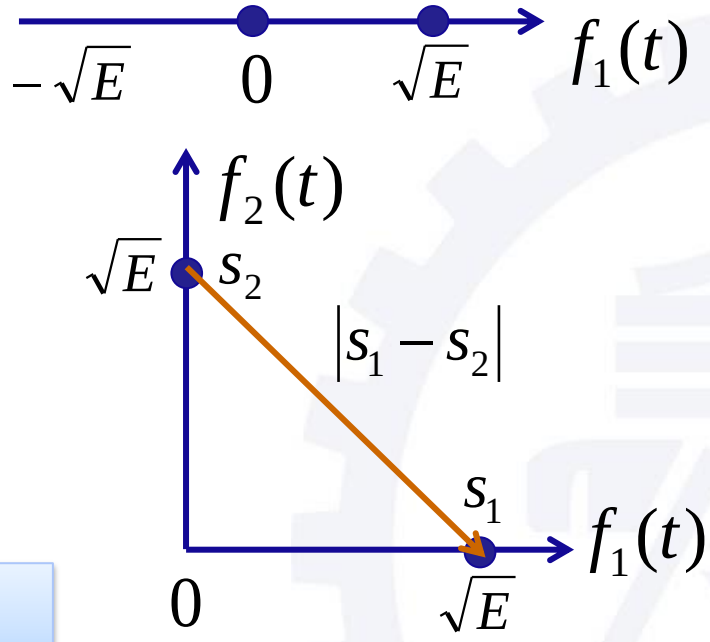
$$P_e = \frac{1}{2}erfc\left(\sqrt{\frac{E_b}{n_0}}\right)$$

- FSK signal

$E_1 = E_2 = E_b = A^2T_b/2,$

$\rho_{12} = 0,\ \ d = \sqrt{2E_b}$

$$P_e = \frac{1}{2}erfc\left(\sqrt{\frac{E_b}{2n_0}}\right)$$



Same E_b/n_0, $P_{e\,PSK} < P_{e\,FSK} < P_{e\,ASK}$

Same P_e, E_b/n_0 of PSK, FSK, ASK are 3dB lower successively

PERFORMANCE ANALYSIS OF OPTIMUM RECEIVER BIT ERROR: SUMMARY

- Optimum receiver bit error rate is determined by signal distance and noise power spectral density. Farther distance, lower bit error rate
- Signal distance is correlated to signal average energy and correlation coefficient